Hyperspectral Image Spectral–Spatial-Range Gabor Filtering

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***Abstract*— Spectral–spatial Gabor filtering, which is based on 3-D local harmonic analysis, has been a powerful spectral–spatial feature extraction tool for hyperspectral image (HSI) classification. However, existing spectral–spatial Gabor approaches are prone to oversmoothing, neglecting the existences of edges and negatively affecting the classification. In this article, we propose a new HSI Gabor filtering concept, called spectral–spatial-range Gabor filtering, which intends to restrain edge interference from disturbing local spectral–spatial harmonic components. Contributions and novelties of our work can be identified as follows: 1) an HSI filtering framework is created, which can accommodate various Gabor filtering procedures and hence offer the potential to guide the design of new Gabor filters; 2) following such a unified filtering framework and taking into consideration both local spectral–spatial harmonic characteristics and range domain variations, we develop a new concept of spectral–spatial-range Gabor filtering; and 3) utilizing this proposed Gabor prototype and elaborating mathematical derivations, we achieve a novel discriminative spectral–spatial-range Gabor filtering method, which can deal with discriminative local harmonics and edge interference simultaneously along the spectral–spatial-range domain, obtaining highly discriminative Gabor features while yielding linear computational complexity. Our novel method is evaluated on four real HSI data sets and achieves excellent performances.**

***Index Terms*— Classification, Gabor filtering, hyperspectral image (HSI), spectral–spatial-range filtering.**

I. Introduction

HYPERSPECTRAL images (HSIs) are capable of capturing accurate spectral signatures of the observed materials with many continuous and narrow bands. This brings new exciting avenues for remotely sensed image classification tasks, which attracted significant attention from the remote sensing community [1]–[4]. In the early development stage of HSI classification techniques, researchers mainly focused on pixel-wise classification, exclusively utilizing spectral signatures to classify pixels [4]. These techniques need to circumvent some challenges. The first one is the spectral variability introduced by many potential interferers, such as incident illumination and shade, atmospheric effects, and instrument noises [5]. These lead to high intraclass variability and low interclass variability. A second challenge results from the high data dimensionality involved in HSIs, which requires a large number of labeled samples to relieve the curse of dimensionality [6], [7]. Such challenges limit the application of pixelwise classifiers in the HSI domain. Therefore, spectral–spatial techniques, which integrate both spectral and spatial information and hence exhibit the potential to significantly enhance the classification, became appealing and predominant in the HSI classification domain [1], [4]. It is noteworthy that the HSI classification technique has been found to be also a key to an emerging field of HSI machine vision including food quality detection, artwork and document investigation, and forensic inspection [8]–[10].

In the literature, spectral–spatial classification methods have been designed with various considerations. Based on the stage where spatial information is exploited, they can be roughly grouped into three categories of: 1) preprocessing-based classification; 2) integrated classification; and 3) postprocessing-based classification methods [1]. In 1), spectral–spatial features are first extracted and then fed to a standard classifier, where spatial information is extracted ahead of classification [1]. In 2), spatial and spectral information are jointly used to establish a unified classifier without explicitly separating the two sources of information [1], [11], [12]. Representative instantiations include the works in [13] and [14], where Laplacian-based and superpixel-based spatial dependencies are leveraged to build joint spare-representation models. In 3), spatial information is utilized to smooth or postregularize some preliminary results obtained by spectral classification, where spatial information works in the later part of the entire spectral–spatial classification process [1]. For example, segmentation, which is able to automatically determine the irregular spatial dependencies in an HSI, has been utilized to build a postspatial regularization for spectral–spatial classification [15], [16].

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Among the three categories aforementioned, preprocessing-based classification has attracted special attention [1], [17] as it possesses two major advantages. The first and foremost is that preprocessing-based classification is conceptually clear. Spectral–spatial features yielded during that classification tend to gain less uncertainty than the original ones. This implies that the newly obtained spectral–spatial feature space is associated with sharper class-specific likelihood functions under the given training samples [18], [19], which result in two important advantages for the late classification: 1) the average curvatures of class-conditional data distributions increase [18]–[20] the meaning that it is easier to achieve a better estimation of the involved parameters—which can be quantitatively measured by Cramer–Rao lower bound [21], [22] and 2) different class-conditional data distributions overlap less with each other, thus reducing error decision risks [7], [23]. Another important advantage is the ease of implementation. Typically, preprocessing-based classification can be easily partitioned into two sequential stages [17], [24]: 1) spectral–spatial feature extraction, used to build representative features and 2) feature-wise classification. For the latter phase, standard classifiers such as support vector machines (SVMs) [25], multinomial logistic regression (MLR) [26], K-nearest neighbors (KNN) [27], and sparse representation (SR)-based classifiers [13], [28] can be conveniently combined with the obtained spectral–spatial features to yield the final classification results.

Apparently, in preprocessing-based classification, spectral–spatial feature extraction has crucial influences on subsequent classification tasks. In fact, spectral–spatial feature extraction has fostered many developments in the last decades [1], [17], of which morphological methods comprise a highly representative group. For example, Benediktsson *et al.* [29], [30], Plaza *et al.* [31], [32], and Fauvel *et al.* [33] employed mathematical morphology operators with structural elements of various sizes to obtain extended morphological profiles for classification purposes. Xu *et al.* [34] extracted multiple morphological components to improve classification performance. Over the past decade, some machine vision techniques have been applied in HSI classification systems, one example is the Gabor methodology [35], [36], which has the potential to extract another representative kind of spectral–spatial features [37]–[40]. Similar to some synthetic aperture radar texture-catching ways such as kurtosis wavelet energy and curvelet coefficient energy [41], [42], Gabor filtering is found to extract the HSI texture characteristic. Before being introduced into HSI classification, Gabor filtering has been successfully used in many different areas, such as biometric recognition [43]–[45], texture segmentation [46], [47], color image segmentation [48], signature recognition [49], and motion tracking [50], which is inspired mainly by two aspects: one is biological interpretation [51], whereas the other is signal processing [35], [52], [53]. Almost all HSI-specific Gabor methods are based on spectral–spatial Gabor filtering, which uses 3-D Gabor filters to capture spectral and spatial information simultaneously. These methods can be categorized as follows.

1) *Regular Spectral–Spatial Gabor Filtering:* In this context, the HSI filtering is basically performed with a standard 3-D Gabor filter, just considering the 3-D data arrangement of HSI cubes. For instance, Bau *et al.* [38] used the real part of 3-DGF to filter hyperspectral images. Shen and Jia [39] exploited both the real and imaginary parts of standard 3-D Gabor filters to extract magnitude features. Jia *et al.* [40] combined such features with least squares (LS) regression-based classifier. However, except for the 3-D data arrangement, other specific HSI characteristics are not included in these methods. That is to say, this kind of methods does not adjust the filter structures to accommodate the discriminative characteristics of HSIs. Additionally, such kinds of filters exhibit cubic computational complexity.
2) *Discriminative Spectral–Spatial Gabor Filtering:* In this category, the discriminative characteristics of HSI data are considered in the design of the filters. One example is the discriminative Gabor filter developed by He *et al.* [37]. In this method, the Gabor kernel was re-designed to accommodate the fact that the discriminative information in HSIs tends to aggregate on low frequencies in the spatial domain (and away from low frequencies in the spectral domain) to improve the discriminative ability of Gabor features.

Existing HSI spatial–spectral Gabor methods, however, tend to over-smooth the images, hence producing spatially over-covered features, as regional boundaries and neighboring classes are not taken into consideration in the method design. It is known that, with 3-D pictorial data arrangement, HSIs usually exhibit spatial discontinuities; that is, some special spatial characteristics such as intensity and texture are sharply adjusted at the regional edges or boundaries. Spectral–spatial Gabor filtering runs with a Gabor kernel sliding across the HSI. However, the spectral–spatial Gabor kernel may span on different class-specific regions when it slides nearby class borders, thus covering pixels from classes different from that of the central pixel. In this situation, the local spectral–spatial harmonic features connected to that central pixel would be polluted by those pixels from other classes, hence reducing the discriminability. In order to address this problem, in this article, we develop a new spectral–spatial-range Gabor filtering concept and an associated filtering method for HSI classification. More specifically, novelties and innovative contributions of this article can be identified as follows.

1) *HSI Filtering Framework:* We establish an HSI filtering framework which reflects adaptations to the local data structure of the available HSI. This framework not only has clear conceptual interpretation but also is general enough to accommodate various Gabor filtering procedures. Thus, it is able to guide the design of new types of Gabor filters. To the best of our knowledge, there is no previous work that explicitly defines an HSI filtering framework such as the one that we are introducing in our work.
2) *Spectral–Spatial-Range Gabor Filtering:* Following our HSI filtering framework, we propose a new Gabor

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filtering concept: spectral–spatial-range filtering, which involves both spectral–spatial harmonic and range considerations. Then, we develop a specific discriminative spectral–spatial-range Gabor filtering method by coupling discriminative spectral–spatial Gabor kernels with a range kernel, which comprises both projecting HSI data onto the spectral–spatial-range kernel and a normalization process. Such a Gabor filtering method, unlike other existing Gabor methods, is able to settle discriminative spatial–spectral harmonic components and sharp spectrum variations along spatial boundaries in an HSI. To the best of our knowledge, all previously existing HSI Gabor methods belong to the spectral–spatial category, and this is the first work that introduces the concept of a spectral–spatial-range Gabor filter, further developing a new spectral–spatial-range Gabor filtering method.

3) *Tractable Spectral–Spatial-Range Implementation:* The spectral–spatial-range Gabor filtering prototype mentioned above is a nonlinear filter with high computational complexity and no frequency-domain interpretation, being in fact computationally intractable in some situations. By means of the sift properties of unit range impulse in the spectral–spatial-range domain and in the spatial–range domain, which can perform dimensional expansion (and taking advantage of a newly developed mathematical process), we introduce a linear separable convolutional implementation of the proposed discriminative spectral–spatial-range Gabor filtering method, which is also an innovative contribution of this article.

The remainder of this article is organized as follows. Section II describes the related works. The proposed Spectral–spatial-range Gabor filtering method is presented in detail in Section III. Section IV demonstrates the effectiveness of our newly developed technique using different HSI data sets. Finally, Section V concludes this article with some remarks and hints at plausible future research lines.

II. Related Work

Existing HSI Gabor filtering methods are usually built on 3-D Gabor filters and belong to the category of spectral–spatial processing. Let (x, y, b) denote the spatial–spectral domain associated with an HSI $\mathbf{h}(x, y, b)$, where (x, y) represents the spatial domain and b stands for the spectral domain. A standard 3-D Gabor kernel can be mathematically represented by a 3-D harmonic modulated by a Gaussian envelope as follows:

$$\begin{aligned}\mathbf{G}_{\boldsymbol{\omega},\theta,\varphi}(x, y, b) = &\frac{1}{(2\pi)^{\frac{3}{2}}\sigma_x\sigma_y\sigma_b} \\ &\times \exp\left\{-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2}+\frac{y^2}{\sigma_y^2}+\frac{b^2}{\sigma_b^2}\right)\right\} \\ &\times \exp\left\{j\left(x\omega_x + y\omega_y + b\omega_b\right)\right\} \qquad (1)\end{aligned}$$

where $\omega_x = |\boldsymbol{\omega}|\sin\varphi\cos\theta$, $\omega_y = |\boldsymbol{\omega}|\sin\varphi\sin\theta$ and $\omega_b = |\boldsymbol{\omega}|\cos\varphi$ are the projections of the given angular frequency $\boldsymbol{\omega}$ onto the x-, y- and b-directions, respectively; φ is the angle between the frequency $\boldsymbol{\omega}$ and the b-direction, and θ is the angle between the projection of $\boldsymbol{\omega}$ onto the (x, y)-plane and the x-direction; $|\boldsymbol{\omega}| = (\omega_x^2 + \omega_y^2 + \omega_b^2)^{(1/2)}$ is the magnitude of $\boldsymbol{\omega}$; and j stands for the imaginary unit.

With the aforementioned 3-D Gabor filter in mind, a standard spatial–spectral Gabor filtering (S^3GF) for an HSI $\mathbf{h}(x, y, b)$ can be formulated with the projection operator g as follows:

$$\begin{aligned}&g\mathbf{h}_{\boldsymbol{\omega},\theta,\varphi}(x, y, b) \\ &\quad= \sum_m\sum_n\sum_l \mathbf{h}(m, n, l)\mathbf{G}_{\boldsymbol{\omega},\theta,\varphi}(m-x, n-y, l-b) \\ &\quad= \mathbf{h}(x, y, b) * \mathbf{G}^c_{\boldsymbol{\omega},\theta,\varphi}(-x, -y, -b) \\ &\quad= \mathbf{h}(x, y, b) * \mathfrak{Re}\{\mathbf{G}_{\boldsymbol{\omega},\theta,\varphi}(-x, -y, -b)\} \\ &\qquad- j\mathbf{h}(x, y, b) * \mathfrak{Im}\{\mathbf{G}_{\boldsymbol{\omega},\theta,\varphi}(-x, -y, -b)\} \qquad (2)\end{aligned}$$

where $*$ denotes the 3-D convolution calculation, the exponent c denotes the conjugation operation, and $\mathfrak{Re}(\cdot)$ and $\mathfrak{Im}(\cdot)$ are the real and the imaginary parts of a 3-D signal, respectively. In order to cover the useful local harmonic information, a set of Gabor filters with different combinations of frequencies and orientations are usually convolved with the HSI. In (2), HSIs are viewed just as a 3-D arrangement on which a standard 3-D Gabor filter is applied, without taking into consideration other special characteristics of HSIs.

Following the formulation in (2), Bau *et al.* [38] computed only the real part of S^3GF to build region-based spectral–spatial features from HSIs. Shen and Jia [39] used the magnitudes of the S^3GF results, obtained from both real and imaginary parts, to serve as spectral–spatial features for HSI classification purposes. Jia *et al.* [40] also utilized the aforementioned Gabor magnitude features for classification purposes, where high-dimensional features were first obtained under different parameter settings and then directly fed into a least squares-based collaborative representation classifier to avoid feature selection.

In the abovementioned Gabor methods, some strategies such as parameter-tuning, generation of amplitude, or energy features, and a combination with suitable postprocessing steps were utilized to improve the discriminability of the Gabor features and hence, improve the classification performance. However, except the 3-D data arrangement, they failed to explicitly involve any HSI characteristics. That is, useful HSI characteristics were not utilized to adjust the inherent Gabor filter structure.

He *et al.* [37] unveiled a latent structure of spectral–spatial Gabor filtering by decomposing the standard 3-D spectral–spatial Gabor filter into the superposition of eight subfilters

$$\begin{aligned}&\mathbf{G}'_{\boldsymbol{\omega},\theta,\varphi}(x, y, b) \\ &= g^{(x)}_{\cos}g^{(y)}_{\cos}g^{(b)}_{\cos} - g^{(x)}_{\cos}g^{(y)}_{\sin}g^{(b)}_{\sin} - g^{(x)}_{\sin}g^{(y)}_{\sin}g^{(b)}_{\cos} - g^{(x)}_{\sin}g^{(y)}_{\cos}g^{(b)}_{\sin} \\ &\quad + j\left\{g^{(x)}_{\sin}g^{(y)}_{\cos}g^{(b)}_{\cos} - g^{(x)}_{\sin}g^{(y)}_{\sin}g^{(b)}_{\sin} + g^{(x)}_{\cos}g^{(y)}_{\sin}g^{(b)}_{\cos} + g^{(x)}_{\cos}g^{(y)}_{\cos}g^{(b)}_{\sin}\right\} \qquad (3)\end{aligned}$$

where $g^{(x)}_{\cos}$, $g^{(y)}_{\cos}$, $g^{(b)}_{\cos}$, $g^{(x)}_{\sin}$, $g^{(y)}_{\sin}$, and $g^{(b)}_{\sin}$ are Gaussian enveloped cosine and sinusoidal harmonics along the $x-$ and $y-$ spatial dimensions and the $b-$ spectral dimension, respectively, combinations of which exhaust all the possible local

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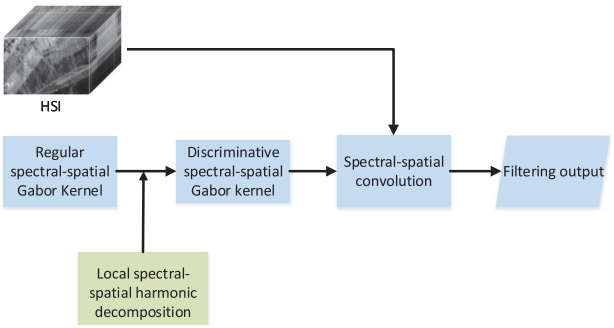


Fig. 1. Illustrative diagram of DS^{2}GF method.

spectral–spatial frequency cases along all the possible dimensions. Furthermore, the frequency-domain characteristics of HSIs are not homogeneous. The discriminative component tends to lie on the low frequencies in the spatial dimension and away from the low frequencies in the spectral dimension. As such, He *et al.* [37] preserved only the subfilter connected to the discriminative component as follows:

$$\mathbf{G}''_{\boldsymbol{\omega},\theta,\varphi}(x,y,b) = g^{(x)}_{\cos} g^{(y)}_{\cos} g^{(b)}_{\sin} \tag{4}$$

and designed a discriminative spectral–spatial Gabor filtering (referred to hereinafter as DS^{2}GF)

$$g_d \mathbf{h}_{\boldsymbol{\omega},\theta,\varphi}(x,y,b) = \mathbf{h} * \mathbf{G}''_{\boldsymbol{\omega},\theta,\varphi}(x,y,b). \tag{5}$$

The design diagram of the above DS^{2}GF is shown in Fig. 1.

III. Spectral–Spatial-Range Gabor Filtering

All previously developed HSI Gabor methods are based on spectral–spatial Gabor filtering. Traditional spectral–spatial Gabor filtering approaches are unable to handle the situation of spatial regional boundaries and adjacent classes, meaning that the Gabor features of class-specific pixels near class regional boundaries may be polluted by those pixels from other classes, hence reducing the discriminability. This section aims at developing a novel spectral–spatial-range Gabor filtering methodology which improves discriminability by considering not only local discriminative frequency components but also spatial regional boundaries and class adjacency. In the following, we first give a general HSI filtering framework which is able to accommodate various kinds of HSI Gabor filters. Then, following such a universal framework, we propose a new concept of spectral–spatial-range Gabor filtering and accordingly design a specific new Gabor filtering method.

A. HSI Filtering Framework

Nonparametric point estimation technique can adapt to the local spatial structure of images, which has pushed the development of modern image filtering methodologies [54]–[56]. Inspired by such an estimation technique, we can formulate HSI filtering as minimizing the following objective:

$$J = \sum_m \sum_n \sum_l \{\{h(m,n,l) - h'(x,y,b)\}^2 \cdot \mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\}\} \tag{6}$$

where $h'(x,y,b)$ is the expected filtering result at spectral–spatial location (x,y,b), $h(m,n,l)$ is the observation at the location (m,n,l), $r(m,n)$ is a range function suggesting an attribute at the spatial location (m,n) which, for instance, can be a quantity reflecting the overall spectral intensity across all the bands at (m,n), and $\mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\}$ is a weight function, also called the kernel function, meaning that the estimation $h'(x,y,b)$ should be influenced by the differences between (m,n,l) and (x,y,b).

The derivative of the objective in (6) is of the form

$$\begin{aligned}\frac{\partial J}{\partial h'(x,y,b)} &= \sum_m \sum_n \sum_l \left\{ \frac{\partial \{h(m,n,l) - h'(x,y,b)\}^2}{\partial h'(x,y,b)} \cdot \mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\} \right\} \\ &= -2 \sum_m \sum_n \sum_l \left\{ \{h(m,n,l) - h'(x,y,b)\} \cdot \mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\} \right\}.\end{aligned} \tag{7}$$

Let $(\partial J / \partial h'(x,y,b)) = 0$, then we have

$$\begin{aligned}h'(x,y,b) &= \frac{1}{v} \sum_m \sum_n \sum_l h(m,n,l) \mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\} \\ &= \frac{1}{v} \langle \mathbf{h}(m,n,l), \mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\rangle_{m,n,l} \\ &= \frac{1}{v(x,y,b)} h''(x,y,b)\end{aligned} \tag{8}$$

where $\langle \cdot,\cdot \rangle_{m,n,l}$ denotes the inner product of two 3-D signals or two 3-mode tensors [57] defined on the domain (m,n,l)

$$\begin{aligned}v(x,y,b) &= \sum_m \sum_n \sum_l \mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\} \\ &= \sum_m \sum_n \sum_l \mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\} \cdot 1 \\ &= \langle \mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\}, \mathbf{1}_3 \rangle_{m,n,l}\end{aligned} \tag{9}$$

is a normalization factor where $\mathbf{1}_3$ is a 3-mode tensor whose entries are all equal to 1, and

$$h''(x,y,b) = \langle \mathbf{h}(m,n,l), \mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\rangle_{m,n,l} \tag{10}$$

is the unnormalized version of the filtering. The normalization process in (8) makes the estimation $h'(x,y,b)$ be surrounded by the observations $h(m,n,l)$ in the spectral–spatial domain, i.e., keeping the filtering result strictly faithful to the observed pixels. This is especially critical for recovery-related tasks such as denoising and deblurring. Equation (8) indicates that the HSI filtering is a procedure able to build linear combinations or affine relationships between HSI pixels of interest

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and the observed HSI pixels, which can also be interpreted as projecting the HSI onto a kernel and, thus, a matched filtering process [20]. More importantly, (8) offers a relatively general or universal HSI filtering framework that we can use to accommodate existing Gabor filters for HSI classification and guide the design of new HSI filtering methods.

Comparing (2) with (8), the standard spectral–spatial Gabor filtering shown in (2) can be viewed as a specific realization of the filtering in (8). It adopts the following kernel function:

$$\mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\} = \mathbf{G}^{c}_{\boldsymbol{\omega},\theta,\varphi}(m-x,n-y,l-b) \quad (11)$$

and cuts out the normalization factor, which means that the filtering is actually reduced to an unnormalized filtering as shown in (10). This is rational because the normalization factor $v(x,y,b)=\sum_m\sum_n\sum_l \mathbf{G}^{c}_{\boldsymbol{\omega},\theta,\varphi}(m-x,n-y,l-b)$ here remains constant over spatial locations (x,y) and, hence, it does not affect the pixel's discriminative ability. Moreover, the application of feature extraction techniques does not require the results of HSI filterings to be strictly faithful to the original observed data. Thus, the normalization factor here can be dropped in the filtering process, which makes the mathematical representation simpler. Then, through inspecting the DS^{2}GF shown in (5), it is found that this filtering is another instantiation of the HSI filtering framework in (8), whereas the kernel function involved is

$$\mathbf{k}_d\{m,n,l,x,y,b,r(m,n),r(x,y)\} = g^{(x)}_{\cos}(m-x)g^{(y)}_{\cos}(n-y)g^{(b)}_{\sin}(l-b) \quad (12)$$

and the normalization factor is also dropped. Similarly, the filtering method associated with the Gabor filter in (3) is also an example of an HSI filtering in (8), where the normalization factor is removed and the kernel function is set to be

$$\mathbf{k}_f\{m,n,l,x,y,b,r(m,n),r(x,y)\} = \mathbf{G}^{'c}_{\boldsymbol{\omega},\theta,\varphi}(m-x,n-y,l-b). \quad (13)$$

Therefore, all of the spectral–spatial Gabor filters related to (2), (3) and (5) can be viewed as instances of the unified HSI filtering framework suggested in (8) with different specifications, where the connected spectral–spatial Gabor kernels are those demonstrated in (11), (12), and (13), respectively.

B. Discriminative Spectral–Spatial-Range Gabor Filtering

By means of the HSI filtering framework presented in (8), we propose a new concept of spatial–spectral–range Gabor filter and then specifically design a connected filtering method which can settle both spatial region boundaries and discriminative local frequency components, while running in a linearly separable convolution manner.

1) Nonlinear Spectral–Spatial-Range Gabor Prototype: As suggested in (8), the nature of an HSI filter is to establish an affine relationship between the expected filtering result and the available observed pixels. In this relationship, the weights of pixels with respect to a pixel of interest are assigned with the kernel $\mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\}$. By closely inspecting (11), (12), and (13), we can find that existing HSI Gabor filters arrange the kernel $\mathbf{k}\{m,n,l,x,y,b,r(m,n),r(x,y)\}$ in the context of local spectral–spatial harmonic analysis under the Euclidean distance. Namely, the expected filtering result for a pixel of interest is only the sum of all the neighboring pixels, weighted by local harmonics in the spectral–spatial domain, without the range domain quantities $r(m,n)$ and $r(x,y)$ taken into consideration. If the range domain is set to be intensity-related, the difference of $r(m,n)$ and $r(x,y)$ can be used to detect the HSI intensity jumping along the spatial regional boundaries [54], [56], [58], thus offering a mechanism to possibly avoid that the local harmonic features of a class-specific pixel are interfered by pixels not in the same class. However, existing HSI Gabor methods neglect such a kind of range considerations and there is no quantity to settle spatial boundaries separating different classes, hence failing to properly capture the real local harmonics near spatial regional boundaries during the filtering process.

Considering the weakness of existing HSI Gabor filterings discussed above, we establish the following Gabor kernel:

$$\mathbf{k}_{g,r}\{m,n,l,x,y,b,r(m,n),r(x,y)\} = \mathbf{k}_g\{m,n,l,x,y,b,r(m,n),r(x,y)\}g^{(r)} \times\{r(m,n)-r(x,y)\} \quad (14)$$

where $\mathbf{k}_g\{m,n,l,x,y,b,r(m,n),r(x,y)\}$ is a spectral–spatial Gabor kernel that captures the local spectral–spatial harmonics of an HSI, and $g^{(r)}(r)=(1/(2\pi)^{(1/2)}\sigma_r)\exp(-(r^2/2\sigma_r^2))$ is a Gaussian kernel in the range domain that captures the attribute differences among different pixels. If the attribute is set to be properly intensity-related, HSI intensity variations along the spatial regional boundaries can adjust the weighted sum process in HSI filtering, thus making the filtering adapt to spatial regional boundaries automatically. For instance, the first principal component of an HSI can be used as a simple range variable to measure the HSI intensity variations over all the bands. As the kernel function shown in (14) involves spectral, spatial, and range factors, thus comprising the spectral, spatial, and range information, we name this kind of Gabor kernel *spectral–spatial-range Gabor kernel*. Following the universal HSI filtering framework in Eq. (8) and the kernel expressing in (14), we can define the following filtering:

$$g_{g,r}\mathbf{h}(x,y,b) = \frac{1}{v_{g,r}(x,y,b)}\langle \mathbf{h}, \mathbf{k}_{g,r} \times\{m,n,l,x,y,b,r(m,n),r(x,y)\}\rangle_{m,n,l} \quad (15)$$

where $v_{g,r}(x,y,b)$ is the normalization factor corresponding to the Gabor projection $\langle \mathbf{h}, \mathbf{k}_{g,r}\rangle_{m,n,l}$. The Gabor filtering shown in (15) is based on spectral–spatial-range Gabor kernels, which jointly include the characteristics of both local spectral–spatial harmonics and range variations, and thus can be called *spectral–spatial-range Gabor filtering*.

In (14), if $\mathbf{k}_g\{m,n,l,x,y,b,r(m,n),r(x,y)\}$ is set to be the standard spectral–spatial Gabor kernel $\mathbf{G}'_{\boldsymbol{\omega},\theta,\varphi}(x,y,b)$ shown in (1), we have

$$g_{s,r}\mathbf{h}(x,y,b) = \frac{1}{v_{s,r}(x,y,b)}\langle \mathbf{h}, \mathbf{k}_{s,r} \times\{m,n,l,x,y,b,r(m,n),r(x,y)\}\rangle_{m,n,l} \quad (16)$$

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where $\upsilon_{s,r}(x, y, b)$ is the normalization factor associated with $\langle \mathbf{h}, \mathbf{k}_{s,r}\rangle_{m,n,l}$. Similarly, if we set $\mathbf{k}_g\{m, n, l, x, y, b, r(m, n), r(x, y)\}$ in (14) to be the discriminative spectral–spatial Gabor kernel shown in (12), we can specifically obtain a filtering representation as follows:

$$\begin{aligned} & g_{d,r}\mathbf{h}(x, y, b) \\ &= \frac{1}{\upsilon_{d,r}(x, y, b)}\langle \mathbf{h}, \mathbf{k}_{d,r}\{m, n, l, x, y, b, r(m, n), r(x, y)\}\rangle_{m,n,l} \\ &= \frac{1}{\upsilon(x, y, b)}\sum_m\sum_n\sum_l\{\mathbf{h}(m, n, l) \\ &\qquad \cdot \mathbf{k}_{f,r}\{m, n, l, x, y, b, r(m, n), r(x, y)\}\} \\ &= \frac{1}{\upsilon_{d,r}(x, y, b)} g'_{d,r}\mathbf{h}(x, y, b) \end{aligned} \tag{17}$$

where

$$\begin{aligned} & \mathbf{k}_{d,r}\{m, n, l, x, y, b, r(m, n), r(x, y)\} \\ &= g^{(x)}_{\cos}(m - x)g^{(y)}_{\cos}(n - y)g^{(b)}_{\sin}(l - b)g^{(r)}\{r(m, n) - r(x, y)\} \end{aligned} \tag{18}$$

is a spectral–spatial-range Gabor kernel which is composed of discriminative harmonics and range variations, and $\upsilon_{d,r}(x, y, b) = \sum_m\sum_n\sum_l \mathbf{k}_{d,r}\{m, n, l, x, y, b, r(m, n), r(x, y)\} = \langle \mathbf{k}_{d,r}, \mathbf{1}_3\rangle_{m,n,l}$ is the associated normalization factor, while

$$g^{(x)}_{\cos} = \frac{1}{(2\pi)^{\frac{1}{2}}\sigma_x}\exp\left(-\frac{x^2}{2\sigma_x^2}\right)\cos(x\omega_x) \tag{19}$$

$$g^{(y)}_{\cos} = \frac{1}{(2\pi)^{\frac{1}{2}}\sigma_y}\exp\left(-\frac{y^2}{2\sigma_y^2}\right)\cos(y\omega_y) \tag{20}$$

$$g^{(b)}_{\sin} = \frac{1}{(2\pi)^{\frac{1}{2}}\sigma_b}\exp\left(-\frac{b^2}{2\sigma_b^2}\right)\sin(b\omega_b). \tag{21}$$

As indicated in (18), if pixels at (x, y) and (m, n) span over a spatial boundary, $|r(m, n) - r(x, y))|$ tends to be larger, $g^{(r)}(r(m, n) - r(x, y))$ tends to be smaller. As a result, the contribution of $h(m, n, l)$ to the filtering result $h(x, y, b)$ is reduced. As the Gabor filtering in (17) demands the availability of discriminative spectral–spatial harmonic components plus the range examination, we call this filtering discriminative spectral–spatial-range Gabor filtering (DS^{2}RGF). As for the filtering mentioned in (16), we call it standard spectral–spatial-range Gabor filtering (S^{3}RGF), as the construction of its spectral–spatial-range Gabor kernel involves the standard Gabor kernel.

Using (15), we define a new concept for HSI Gabor filtering, i.e., spectral–spatial-range Gabor filtering (S^{2}RGF), which reflects not only the consideration of discriminative local harmonic components constituting an HSI but also those of spatial regional boundaries. However, the original S^{2}RGF prototype, as well as its two instantiations S^{3}RGF and DS^{2}RGF, incurs problems. Basically, the prototypes in (15), (17) and (16) define HSI filtering processes, but they are nonlinear 3-D filters. Take DS^{2}RGF prototype as an example. When the kernel $\mathbf{k}_{d,r}\{m, n, l, x, y, b, r(m, n), r(x, y)\}$ moves across the spectral–spatial domain (x, y, b), its value does not keep invariant because the range values vary over (x, y) if the range domain is set to be the HSI intensities. That is, the value of the kernel in (18) depends on its spatial location (x, y), thus this filtering in (17) is a nonlinear filter that cannot be mathematically represented in a convolution manner. In the real computation of (17), the 3-D kernel shown in (18) has to be updated for every (x, y), and thus (17) cannot be directly decomposed into 1-D convolutions. Therefore, this original DS^{2}RGF prototype does not exhibit linear computation complexity, and would require heavy computational burden in practice. Moreover, due to the nonlinearity of the spectral–spatial-range filtering prototype, we cannot analyze it in the frequency-domain.

In the following, we will focus on the DS^{2}RGF prototype to develop a linearly tractable DS^{2}RGF, as DS^{2}RGF accommodates not only the spectral–spatial-range factor but also the specific discriminative harmonic considerations, which offer the potential to reach better discriminative performance.

2) Dimension Expansion to Spectral–Spatial-Range Domain: Consider the nonlinear prototype in (17). Introducing an extra variable is a widely used strategy to construct a linear data mapping [58], [59]. Inspired by this, we first develop the following relationships about kernel $\mathbf{k}_{d,r}\{m, n, l, x, y, b, r(m, n), r(x, y)\}$:

$$\begin{aligned} & \sum_{r'}\mathbf{k}_{d,r}\{m, n, l, x, y, b, r', r(x, y)\}\delta_r(r' - r(m, n)) \\ &\qquad = \mathbf{k}_{d,r}\{m, n, l, x, y, b, r(m, n), r(x, y)\} \end{aligned} \tag{22}$$

where $\delta_r(\cdot)$ denotes the range impulse $\delta_r(r - r_0) = \begin{cases} 1, & r = r_0 \\ 0, & r \neq r_0 \end{cases}$. Equation (22) introduces a range variable r' independent of the spatial–spectral domain and building a summation along it, together with the summations along the spatial variables m and n, and the spectral variable l. As shown in (17), such a summation along the range variable is expected to help establish a convolution along the spectral–spatial domain. Equation (22) is in fact the sift ability of range impulse along the spectral–spatial-range domain. In addition, we can take advantage of the following relationship related to the spectral–spatial-range domain:

$$\begin{aligned} g'_{d,r}\mathbf{h}(x, y, b) &= g'_{d,r}\mathbf{h}(x, y, b, r(x, y)) \\ &= g'_{d,r}\mathbf{h}(x, y, b, r)|_{r=r(x,y)} \end{aligned} \tag{23}$$

where the quantity $g'_{d,r}\mathbf{h}(x, y, b)$ with three independent variables is viewed as an instance of $g'_{d,r}\mathbf{h}(x, y, b, r)$ with four independent variables, under setting the variable r to be the value $r(x, y)$.

3) Spectral–Spatial-Range Gabor Convolution: With the dimension expansion strategy mentioned above, we can obtain the mathematical derivation shown in (24), which transforms the nonlinear computation in (17) into a separable spectral–spatial-range convolution

$$\begin{aligned} & g'_{d,r}\mathbf{h}(x, y, b, r) \\ &= \left\langle \mathbf{h}, \mathbf{k}'_r\{m, n, l, x, y, b, r(m, n), r\}\right\rangle_{m,n,l} \\ &= \left\langle \mathbf{h}, \sum_{r'}\mathbf{k}'_r\{m, n, l, x, y, b, r', r\}\delta_r(r' - r(m, n))\right\rangle_{m,n,l} \end{aligned}$$

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$$
\begin{aligned}
&= \sum_{l}\sum_{n}\sum_{m}\sum_{r'} \mathbf{h}(m,n,l)\cdot g_{\cos}^{(x)}(m-x)\\
&\quad \cdot g_{\cos}^{(y)}(n-y)\cdot g_{\sin}^{(b)}(l-b)\cdot g^{(r)}(r'-r)\delta_r(r'-r(m,n))\\
&= \sum_{m}\sum_{n}\sum_{r'} g_{\cos}^{(x)}(m-x)\cdot g_{\cos}^{(y)}(n-y)\\
&\quad \cdot g^{(r)}(r'-r)\delta_r(r'-r(m,n))\cdot \sum_{l}\mathbf{h}(m,n,l)\cdot g_{\sin}^{(b)}(l-b)\\
&= \sum_{m}\sum_{n} g_{\cos}^{(x)}(m-x)\cdot g_{\cos}^{(y)}(n-y)\\
&\quad \cdot \sum_{r'} g^{(r)}(r'-r)\delta_r(r'-r(m,n))\cdot\left\{\mathbf{h}(m,n,b) * g_{\sin}^{(b)}(-b)\right\}\\
&= \sum_{m}\sum_{n} g_{\cos}^{(x)}(m-x)\cdot g_{\cos}^{(y)}(n-y)\\
&\quad \cdot\left\{\sum_{r'} g^{(r)}(r'-r)\cdot\delta_r(r'-r(m,n))\{\mathbf{h}(m,n,b) * g_{\sin}^{(b)}(-b)\}\right\}\\
&= \sum_{m}\sum_{n} g_{\cos}^{(x)}(m-x)\cdot g_{\cos}^{(y)}(n-y)\\
&\quad \cdot\left\{g^{(r)}(r) * \{\delta_r(r-r(m,n))\{\mathbf{h}(m,n,b) * g_{\sin}^{(b)}(-b)\}\}\right\}\\
&= \sum_{m} g_{\cos}^{(x)}(m-x)\cdot g_{\cos}^{(y)}(y)\\
&\quad * g^{(r)}(r) * \{\delta_r(r-r(m,y))\{\mathbf{h}(m,y,b) * g_{\sin}^{(b)}(-b)\}\}\\
&= g_{\cos}^{(x)}(x) * g_{\cos}^{(y)}(y) * g^{(r)}(r)\\
&\quad *\{\delta_r(r-r(x,y))\{\mathbf{h}(x,y,b) * g_{\sin}^{(b)}(-b)\}\}\\
&= \mathbf{h}'(x,y,r,b) * g_{\cos}^{(x)}(x) * g_{\cos}^{(y)}(y) * g^{(r)}(r). \qquad (24)
\end{aligned}
$$

In (24), we have

$$
\begin{aligned}
\mathbf{h}'(x,y,r,b) &= \left\{\delta(r-r(x,y))\left\{\mathbf{h}(x,y,b) * g_{\sin}^{(b)}(-b)\right\}\right\}\\
&= \begin{cases}\mathbf{h}(x,y,b) * g_{\sin}^{(b)}(-b) & r = r(x,y)\\ 0 & \text{otherwise.}\end{cases} \qquad (25)
\end{aligned}
$$

By close inspection of (23), (24) and (25), we find that, for a given band b_0, $\mathbf{h}(x,y,r(x,y),b_0)$ is a 3-D data set, and we can compute it bandwise with

$$
\begin{aligned}
&g_{d,r}'\mathbf{h}(x,y,b_0)\\
&\quad = \mathbf{h}'(x,y,r,b_0) * g_{\cos}^{(x)}(x) * g_{\cos}^{(y)}(y) * g^{(r)}(r)|_{r=r(x,y)}. \qquad (26)
\end{aligned}
$$

Then, considering the normalization factor $v_{d,r}(x,y,b)$ in (17), we have

$$
\begin{aligned}
&v_{d,r}(x,y,b)\\
&\quad = \langle \mathbf{k}_{d,r}, \mathbf{1}_3\rangle_{m,n,l}\\
&\quad = \sum_{n}\sum_{m}\sum_{l}\left\{g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g_{\sin}^{(b)}(l-b)\right.\\
&\qquad\qquad \left.\cdot g^{(r)}(r(m,n)-r(x,y))\right\}\\
&\quad = \sum_{l} g_{\sin}^{(b)}(l-b)\sum_{n}\sum_{m}\left\{g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)\right.\\
&\qquad\qquad \left.\cdot g^{(r)}(r(m,n)-r(x,y))\right\}\\
&\quad = \sum_{n}\sum_{m}\left\{g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)\right.\\
&\qquad\qquad \left.\cdot g^{(r)}(r(m,n)-r(x,y))\right\}\cdot\sum_{l} g_{\sin}^{(b)}(l-b). \qquad (27)
\end{aligned}
$$

Observing (27), we can find that the value of $v_{d,r}(x,y,b)$ varies over the spatial location (x,y). Therefore, unlike the Gabor filters shown in (2), (3), and (5), where the normalization factor $v(x,y,b)$ can be totally neglected, here we have to consider the normalization factor in a more subtle manner. Obviously, $\sum_b g_{\sin}^{(l)}(l-b)$ keeps constant over (x,y), having no influence on the pixel's discriminative ability of the filtering result, yielded with (17), and thus can be dropped from the normalization factor. However, the quantity $\sum_n\sum_m\{g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}(r(m,n)-r(x,y))\}$ is dependent on (x,y). Then, we need to calculate only the following simplified normalization factor:

$$
\begin{aligned}
&v_{d,r}(x,y)\\
&\quad = \sum_{n}\sum_{m} g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}(r(m,n)-r(x,y)) \qquad (28)
\end{aligned}
$$

to replace the normalization term $v_{d,r}(x,y,b)$ in (17).

Similar to (23) and (24), we have

$$
\begin{aligned}
&\sum_{r'} g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}(r'-r)\delta_r(r'-r(m,n))\\
&\qquad = g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}(r(m,n)-r) \qquad (29)
\end{aligned}
$$

which is the spatial–range sift ability of range impulse, and there is the relationship

$$
\begin{aligned}
&v_{d,r}(x,y)\\
&= v_{d,r}(x,y,r(x,y))\\
&= v_{d,r}(x,y,r)|_{r=r(x,y)}\\
&= \sum_{n}\sum_{m} g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}(r(m,n)-r)|_{r=r(x,y)}. \qquad (30)
\end{aligned}
$$

Then, it is obtained

$$
\begin{aligned}
&v_{d,r}(x,y,r)\\
&= \sum_{n}\sum_{m} g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}(r(m,n)-r)\cdot 1\\
&= \left\langle g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}(r(m,n)-r), \mathbf{1}_2\right\rangle_{m,n}\\
&= \left\langle \sum_{r'} g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}\right.\\
&\qquad \left.\times (r'-r)\delta_r(r'-r(m,n)), \mathbf{1}_2\right\rangle_{m,n}\\
&= \left\langle g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}(r'-r)\delta_r(r'-r(m,n)), \mathbf{1}_3\right\rangle_{m,n,r'}\\
&= \sum_{n}\sum_{m}\sum_{r'} g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}\\
&\quad \times (r'-r)\delta_r(r'-r(m,n))\\
&= \sum_{n}\sum_{m} g_{\cos}^{(x)}(m-x)g_{\cos}^{(y)}(n-y)g^{(r)}(r) * \delta_r(r-r(m,n))\\
&= g_{\cos}^{(x)}(x) * g_{\cos}^{(y)}(y) * g^{(r)}(r) * \delta_r(r-r(x,y)) \qquad (31)
\end{aligned}
$$

where $\mathbf{1}_2$ is a 2-mode tensor (a matrix) whose entries are all 1, $\mathbf{1}_3$ is a 3-mode tensor with all of its entries being also 1.

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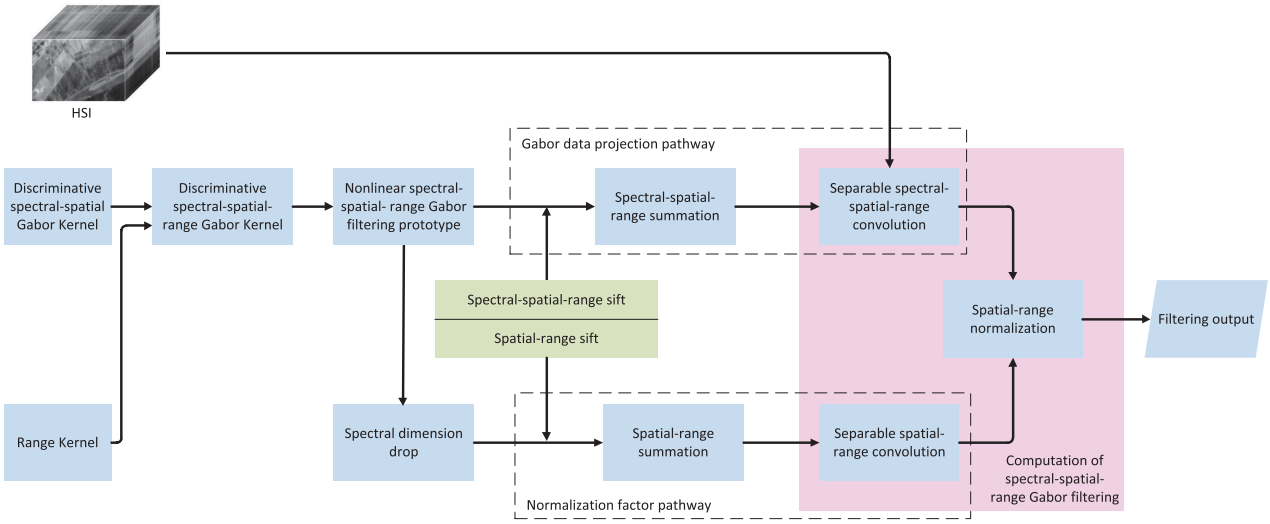


Fig. 2. Illustrative diagram of DS^{2}RGF method.

Algorithm 1 Discriminative Spectral–Spatial-Range Gabor Filtering (DS^{2}RGF)

1) Input: an HSI $\mathbf{h}(x, y, b)$.
2) Compute 1-D filters: $g_{\cos}^{(x)}$, $g_{\cos}^{(y)}$ and $g_{\sin}^{(b)}$ using Equations (19), (20) and (21); and compute the range domain image $r(x, y)$ and then $g^{(r)}$.
3) Compute the convolution

$$\mathbf{h}_b(x, y, b) = \mathbf{h}(x, y, b) * g_{\sin}^{(b)}(-b). \tag{33}$$

4) Compute the normalization factor $\upsilon(x, y)$ using (32).
5) for $b_0 = 1 : B$
   a) Compute $g'_{d,r}\mathbf{h}(x, y, b_0)$ using (26).
   c) Compute $g_{d,r}\mathbf{h}(x, y, b_0)$ using (17).
   end

Combining (30) with (31), we have the normalization factor

$$\upsilon_{d,r}(x, y) = g_{\cos}^{(x)}(x) * g_{\cos}^{(y)}(y) * g^{(r)}(r) * \delta(r - r(x, y))|_{r=r(x,y)}. \tag{32}$$

Substituting (26) and (32) into (17) realizes the linear computation of DS^{2}RGF defined in (17). Fig. 2 shows a design diagram for the proposed DS^{2}RGF method. Our DS^{2}RGF takes into account HSI spatial boundary characteristics and discriminative spectral–spatial local harmonic components, thus improving the HSI Gabor discriminability. Algorithm 1 gives a detailed implementation.

As mentioned in Section III-B1, S^{2}RGF prototypes are nonlinear filters which are unable to run with the convolutional operation. They exhibit cubic computation complexity, making their implementation difficult in real applications, and cannot yield good interpretation in the frequency domain. In contrast, our DS^{2}GF method is a linear filter with 1-D convolutional operations, being able to achieve the linear computational complexity and offering clear filter interpretation in the frequency domain. Moreover, the involved 1-D filters $g_{\cos}^{(x)}(-x)$, $g_{\cos}^{(y)}(-y)$, and $g_{(r)}(-r)$ are all low-frequency pass and are used in DS^{2}RGF. Therefore, the downsampling trick [58], [60] can be used in spatial and range domains when performing DS^{2}RGF, which means that DS^{2}RGF exhibits the potential to further reduce its computational burden in practice.

C. Innovations With Respect to a Previous Work

Here, we specifically conclude the major differences between the proposed DS^{2}RGF method and the existing DS^{2}GF [37] method mentioned in II, thus highlighting the innovations of the proposed work with respect to that previous one. Their differences lie on three major points as follows.

1) The situations that they are suitable for are different. The existing DS^{2}GF in Section II aims to design the HSI Gabor filtering which can accommodate two specific characteristics of HSIs, i.e., 3-D data arrangement and discriminative local harmonic components. Different from that previous work, the proposed DS^{2}RGF is a new kind of Gabor filter capable of automatically adjusting to regional boundaries in HSIs, while not missing the aforementioned two properties of 3-D data arrangement and discriminative harmonic components.
2) Our newly designed filtering is based on different Gabor concepts. What was concerned in Section II, including DS^{2}GF, is the spectral–spatial Gabor filtering methodology, which is defined in the spectral–spatial domain (which mathematically involves only spectral and spatial variables). Different from the previous work, in this article, we develop a new concept for Gabor filtering, the spectral–spatial-range Gabor filtering, which is performed in the spectral–spatial-range domain (mathematically involves not only spectral and spatial variables but also the range variable).
3) Their final implementations and the ways they are derived are totally different. The existing DS^{2}GF uses a harmonic decomposition trick to derive a spectral–spatial Gabor filtering method, which is realized by a

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TABLE I

OAs (%) ALONG WITH STANDARD DEVIATIONS FOR THE AVIRIS INDIAN PINES SCENE, OBTAINED FROM 10 MONTE CARLO RUNS BY THE MLR AND SVM CLASSIFIERS WITH DIFFERENT NUMBERS OF TRAINING SAMPLES PER CLASS, RESPECTIVELY

Classifier	MLR					SVM				
♯Train/class	2	5	8	10	15	2	5	8	10	15
EMAPs	51.01±5.58	70.75±3.66	76.56±2.00	78.77±2.16	84.21±2.62	34.19±8.67	68.63±8.59	77.09±3.56	80.37±1.50	84.94±2.61
S^3GF	42.41±5.86	57.92±3.79	68.80±2.11	71.59±2.03	76.89±1.94	23.02±7.09	57.44±3.11	64.96±3.88	67.53±3.82	75.10±1.81
S^3RGF	51.69±5.88	75.22±2.55	80.21±2.93	81.04±1.94	86.76±2.89	32.18±6.07	71.63±6.38	81.44±2.33	81.78±3.46	86.57±2.75
DS^2GF	46.78±4.87	63.79±3.11	73.24±2.67	75.70±2.32	80.87±1.82	23.10±7.29	60.19±4.62	67.25±3.10	72.29±2.06	78.25±2.01
DS^2RGF	**55.80±4.14**	**75.24±2.43**	**80.55±2.31**	**82.61±1.75**	**87.12±1.95**	**36.26±10.09**	**77.83±2.32**	**81.80±1.88**	**85.27±2.32**	**88.12±2.69**

TABLE II

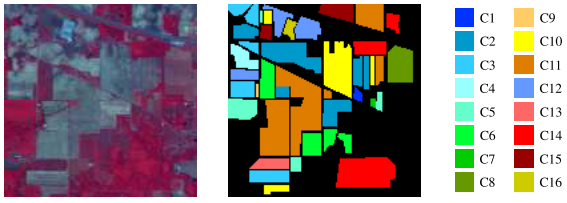
OAs (%) ALONG WITH STANDARD DEVIATIONS FOR THE AVIRIS INDIAN PINES SCENE, OBTAINED FROM 10 MONTE CARLO RUNS BY THE MLR AND SVM CLASSIFIERS WITH DIFFERENT PERCENTAGES OF TRAINING SAMPLES PER CLASS, RESPECTIVELY

Classifier	MLR					SVM				
Sample size(%/class)	1	2	3	4	5	1	2	3	4	5
EMAPs	79.70±2.46	87.35±1.41	90.71±1.53	92.23±0.60	93.48±0.80	75.00±3.27	86.49±1.28	89.65±0.87	91.60±0.95	92.70±0.85
S^3GF	69.59±2.05	81.32±0.87	87.57±0.71	90.22±0.95	92.81±0.76	62.34±3.34	80.30±2.00	87.48±1.10	89.55±1.34	91.46±0.79
S^3RGF	83.03±2.31	90.37±0.93	93.30±0.90	94.78±0.50	95.49±0.45	74.88±2.43	88.14±1.52	92.30±1.06	94.24±0.72	95.47±0.46
DS^2GF	75.49±2.50	85.95±1.34	91.09±1.27	93.21±0.77	94.74±0.51	67.48±3.10	82.96±1.27	88.48±1.05	90.17±0.65	91.88±0.56
DS^2RGF	**83.36±1.91**	**90.68±1.21**	**94.05±0.67**	**94.86±0.68**	**95.77±0.78**	**77.83±2.37**	**89.32±1.41**	**93.06±0.76**	**94.49±0.76**	**95.69±0.72**

linear spectral–spatial convolution (the design diagram is shown in Fig. 1). As opposed to that, DS^2RGF faces a more complex situation. As the proposed spectral–spatial-range Gabor filtering prototype is originally nonlinear, it is computationally intractable and has no frequency-domain interpretation. Thus, we employ mathematical tools such as the sift ability of the range impulse in the spectral–spatial-range domain and that in the spatial–range domain to carry out a dimension expansion and then elaborate further mathematical derivations to transform the filtering so that it now involves only 1-D convolutions (as shown in Fig. 2).

IV. EXPERIMENTAL RESULTS

In this section, we use four real HSIs, i.e., the Indian Pines, Zaoyuan, Salinas, and University of Houston scenes, to evaluate the proposed spectral–spatial-range Gabor filtering methods, DS^2RGF and S^3RGF. For comparison, we also considered the extended morphological attribute profiles (EMAPs) [30], S^3GF and DS^2GF [37] in our experiments. Specifically, the EMAPs were built using threshold values varying from 2.5% to 10% in an interval of 2.5% corresponding to the mean of the individual features and were chosen for the definition of the criteria based on the standard deviation attribute, with 200, 500, and 1000 as references for the area attribute [61]. For Gabor filtering methods, we set ω as $\pi/16$, $\pi/8$, $\pi/4$, and $\pi/2$, while both ϕ and θ as 0, $\pi/4$, $\pi/2$, and $3\pi/4$. Since subfilterings with different θs are the same when ϕ equals 0 for a certain kind of Gabor filters, we finally obtain 52 features corresponding to each band. Afterward, two classifiers, i.e., the MLR and the SVM, are applied to featurewise classification. We utilize the overall accuracy (OA), average accuracy (AA), class individual accuracy (CA), and kappa statistic (κ) as quantitative indicators. For the Indian Pines, Zaoyuan, and Salinas scenes, we randomly selected training samples from the available groundtruth, and used the rest for the test. Considering the statistical significance, we conducted 10 independent Monte Carlo runs and reported the average results. For the University of Houston scene, we used the publicly available training and test sets.



(a) (b)

Fig. 3. AVIRIS Indian Pines scene. (a) False color map. (b) Ground-truth map.

A. Data Set 1: Indian Pines

The first scene was gathered by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor over the Indian Pines test site in Northwestern Indiana, America, available at the Purdue's university MultiSpec site.[1] Although most of the scene is covered by agricultural features, there are still several roads and other built structures like housing. This scene comprises 145 × 145 pixels with a spectral resolution of 10 nm and 224 spectral channels within the wavelength range from 0.4 μm to 2.5 μm. After removing the bands of water absorption, there are totally 200 bands left. The groundtruth available is designated into 16 mixed classes including Alfalfa, Corn-no till, Corn-min till, Corn, Grass/Pasture, Grass/Trees, Grass/Pasture-mowed, Hay-windrowed, Oats, Soybeans-no till, Soybeans-min till, Soybeans-clean till, Wheat, Woods, Buildings-Grass-Tree-Drives, and Stone-steel-towers corresponding to C1–C16 in Fig. 3, where the background is shown in dark color.

[1] https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

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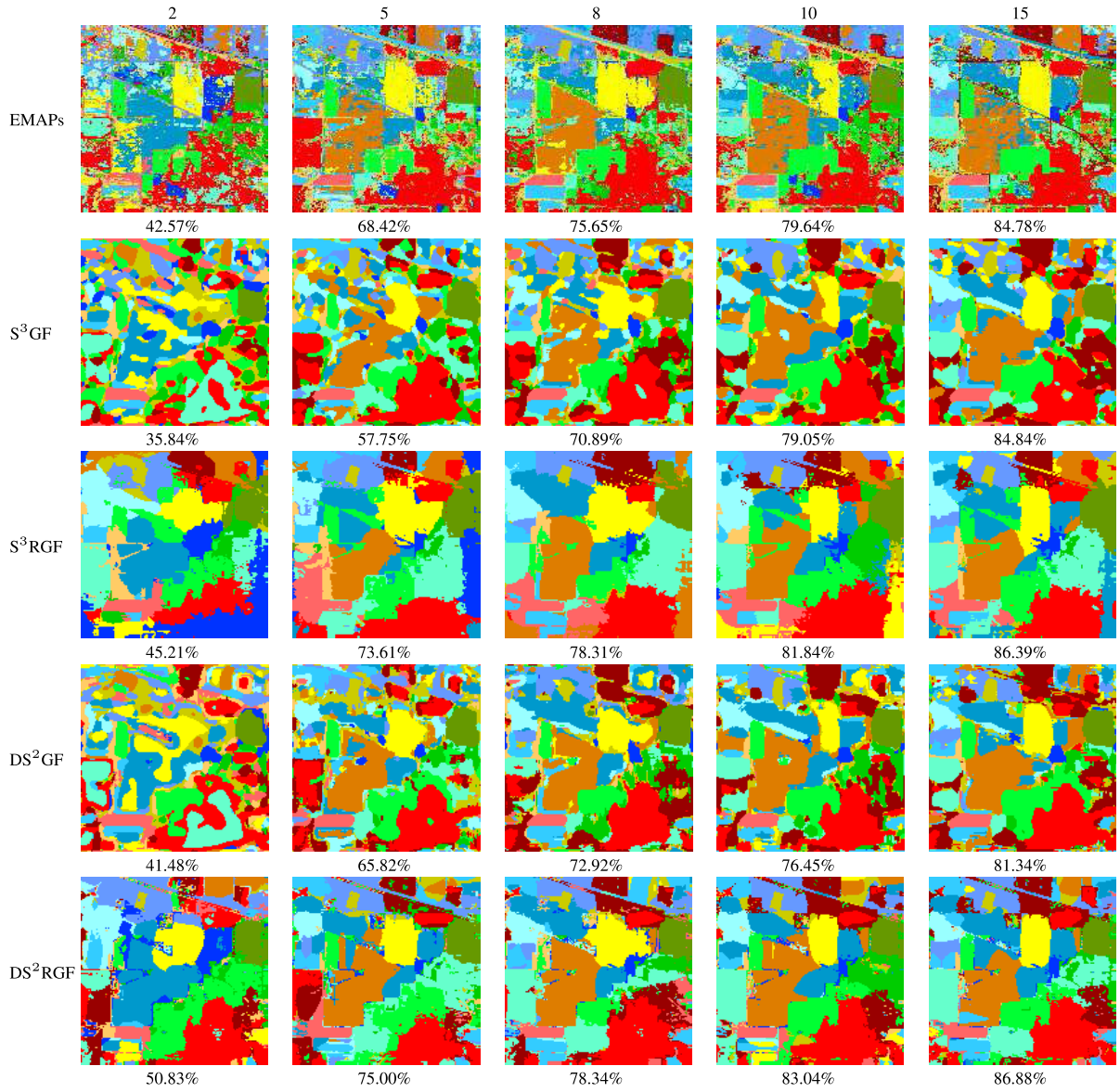


Fig. 4. Classification maps, along with the corresponding OAs, for the AVIRIS Indian Pines scene, obtained from one of the 10 Monte Carlo runs by the MLR classifier with different numbers of training samples per class (corresponding to each column) on the considered features (corresponding to each row).

First of all, we statistically evaluated the classification performance on S^{2}RGF features. For all the test classes, we randomly select an equal number of pixels from each class for training and use the remaining for testing, which is a protocol also adopted in [1], [3], [63], and [64]. Tables I and II list the OAs (along with the standard deviations) obtained by different feature extraction methods with different numbers and percentages of training samples per class, respectively. The latter, selecting an equal percentage of pixels per class for training, is a setting employed in [37], [64], [65]. It can be seen that both S^{3}RGF and DS^{2}RGF, with the sharp spectrum variations along spatial boundaries taken into consideration, obtained very competitive results with regard to their traditional counterparts and EMAPs. Noticeably, the proposed DS^{2}RGF even performed better than S^{3}RGF. The advantages are particularly distinct when very limited training samples were used. This is expected, since DS^{2}RGF is designed to settle discriminative spatial–spectral harmonic components except for the consideration of spatial boundaries in the HSI. For additional details, we report the CAs, OAs, AAs, and κs by MLR with eight training samples per class in Table III. The improvement of S^{2}RGF methods can also be observed from the individual class accuracies, especially on big classes with a large number of samples, such as C2, C11, and C14.

Fig. 4 shows the classification maps obtained by MLR on the considered features with different numbers of training samples. It is remarkable that the classification maps obtained with S^{3}RGF and DS^{2}RGF, i.e., those listed in the third and the last rows, are more like the groundtruth. They are smooth, yet without a severe loss of details. Also, the maps generated by DS^{2}RGF look slightly better than those generated with S^{3}RGF. On the contrary, the maps generated by EMAPs are affected by noises, while those by S^{3}GF and DS^{2}GF are a bit oversmoothed. Furthermore, the ground objects of large

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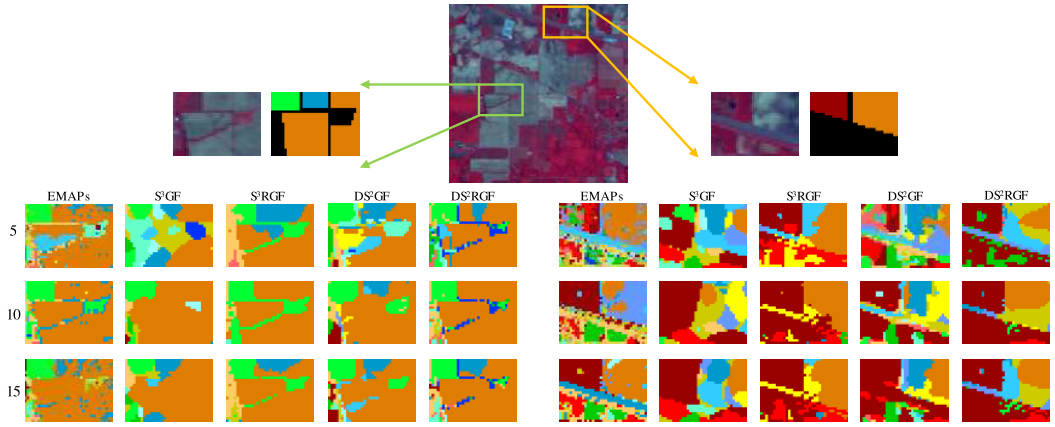


Fig. 5. Two sets of zoomed-in view figures clipped from the classification maps in Fig. 4 for the AVIRIS Indian Pines scene.

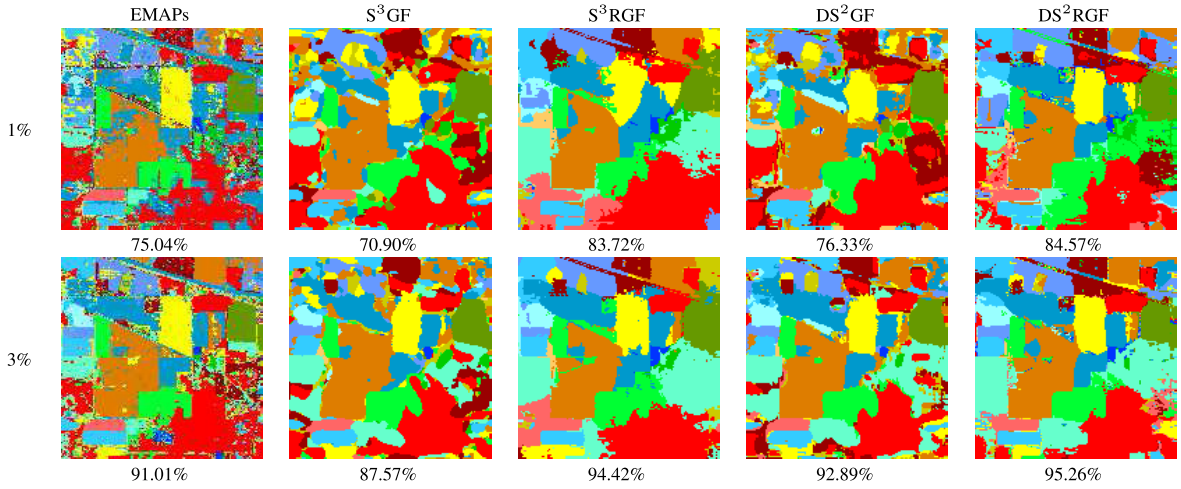


Fig. 6. Classification maps, along with the corresponding OAs, for the AVIRIS Indian Pines scene, obtained from one of the 10 Monte Carlo runs by the MLR classifier with different percentages of training samples per class (corresponding to each row) on the considered features (corresponding to each column).

TABLE III

CAs (%), OAs (%), AAs (%), AND κs (%) FOR THE AVIRIS INDIAN PINES SCENE, OBTAINED AFTER 10 MONTE CARLO RUNS BY THE MLR CLASSIFIER WITH 8 TRAINING SAMPLES PER CLASS

Class	♯Test	EMAPs	S^{3}GF	S^{3}RGF	DS^{2}GF	DS^{2}RGF
C1	46	96.30	96.74	**100.0**	95.22	97.17
C2	1426	61.06	53.49	66.60	67.01	**73.55**
C3	826	70.17	60.44	68.97	65.18	**72.82**
C4	226	83.98	89.56	**95.80**	93.27	95.40
C5	489	71.70	69.43	73.68	**78.47**	75.69
C6	739	**95.18**	84.57	87.08	91.54	85.06
C7	18	99.44	99.44	**100.0**	**100.0**	98.33
C8	481	**98.98**	96.01	94.93	96.47	98.34
C9	12	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
C10	960	77.35	62.24	**79.49**	64.46	79.45
C11	2460	73.79	61.18	**78.84**	62.19	77.26
C12	606	69.72	51.42	78.15	60.13	**78.73**
C13	204	99.12	97.55	99.07	**99.46**	97.11
C14	1286	80.01	81.73	**88.38**	82.59	84.94
C15	372	80.05	88.60	**91.83**	89.60	87.23
C16	87	97.24	**97.70**	95.52	97.01	91.49
OA		76.56	68.80	80.21	73.24	**80.55**
AA		84.63	80.63	87.40	83.91	**87.04**
κ		73.63	65.00	77.70	70.07	**78.07**

size are better recognized in the maps generated by S^{2}RGF, indicating that S^{2}RGF methods are more effective for those large objects as they can remove the noises inside them and simultaneously avoid damaging their boundaries. The same conclusions can be drawn from the comparison of the zoomed-in details illustrated in Fig. 5, where the pavements and trails between or within the fields are more clearly captured by DS^{2}RGF and S^{3}RGF. Furthermore, we list some of the classification maps obtained with different percentages of training samples per class in Fig. 6, which graphically demonstrates the effectiveness of our proposed methods as well.

Next, we conducted principal component analysis (PCA) on the original data, spectral–spatial features, and spectral–spatial-range features and list their first components in Fig. 7. As shown in Fig. 7(d) and (f), the features by DS^{2}RGF and S^{3}RGF stand out with less noises and well-preserved spatial boundaries. In this case, the features by DS^{2}RGF appear to exhibit higher quality, with more enhanced intensity contrast due to the discriminative spectral–spatial local harmonic components. This explains their good performance in the classification tasks. In turn, EMAPs and S^{2}GF methods tend to under and oversmooth the original data, respectively. Nevertheless, all the spectral–spatial and spectral–spatial-range features are more discriminative than the original ones, thus making them more powerful in classification than the original spectral ones.

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TABLE IV

OAs (%) ALONG WITH STANDARD DEVIATIONS FOR THE OMIS ZAOYUAN SCENE, OBTAINED FROM 10 MONTE CARLO RUNS USING THE MLR AND SVM CLASSIFIERS WITH DIFFERENT NUMBERS OF TRAINING SAMPLES PER CLASS, RESPECTIVELY

Classifier	MLR					SVM				
♯Train/class	5	10	15	20	25	5	10	15	20	25
EMAPs	71.36±4.62	82.70±2.03	84.66±1.60	86.82±2.51	87.73±1.15	64.91±5.63	80.47±2.48	84.19±1.31	86.29±1.56	87.82±1.12
S^3GF	71.24±4.47	81.33±1.96	82.79±1.32	85.62±2.06	87.50±1.36	70.40±6.18	82.54±2.45	85.71±1.52	87.65±2.16	89.07±1.39
S^3RGF	77.05±5.89	86.13±2.47	87.51±1.89	90.69±1.30	92.28±1.63	74.08±6.49	85.18±2.43	86.93±1.78	90.06±1.99	90.51±1.25
DS^2GF	71.84±5.21	82.14±3.33	85.20±2.08	86.88±2.32	89.49±1.72	73.31±5.17	83.53±2.99	87.46±1.57	89.43±1.65	90.81±1.36
DS^2RGF	**80.03±4.13**	**87.28±2.41**	**90.64±1.53**	**91.67±1.05**	**92.79±1.23**	**78.37±5.31**	**87.44±2.58**	**89.79±1.93**	**90.98±1.92**	**92.10±1.74**

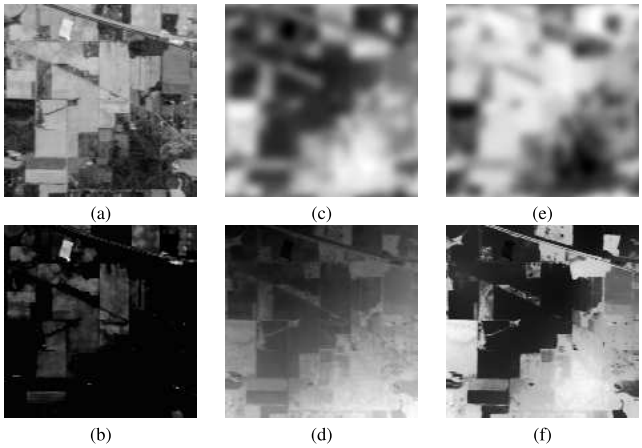


Fig. 7. First principal components obtained on different features for the AVIRIS Indian Pines scene. (a) Original. (b) EMAPs. (c) S^3GF. (d) S^3RGF. (e) DS^2GF. (f) DS^2RGF.

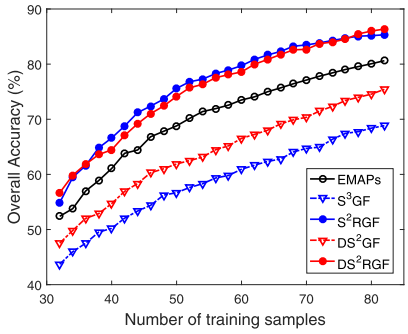


Fig. 8. OAs as a function of the number of training samples, using 32 initial samples (2 per class, 16 classes) and 50 actively selected samples for the AVIRIS Indian Pines scene.

In accordance with [24] and [67], effective features can make a significant improvement in active learning (AL) by reforming data distributions. Thus, we conducted AL and plotted the OAs obtained on different features in Fig. 8. Here, we randomly selected two random training samples from each class, and then added a new one using the *breaking ties* (BT) criterion [67] at each iteration. As illustrated in Fig. 8, the S^2RGF method outperforms the others at each iteration, and gradually improves the OA as more training samples are included. For example, the improvement in OA between DS^2RGF and DS^2GF increases from 9% to 13% during the iterative process. This further verifies the effectiveness of the proposed DS^2RGF method.

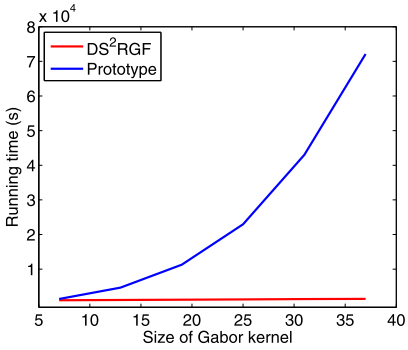


Fig. 9. Computational time as a function of the Gabor kernel size for the AVIRIS Indian Pines scene.

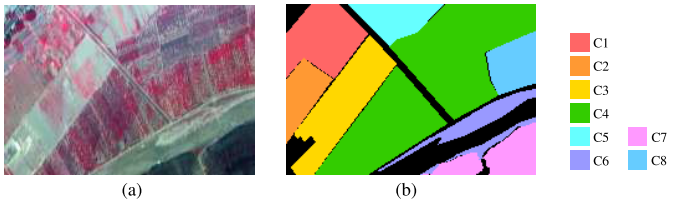


Fig. 10. OMIS Zaoyuan scene. (a) False color map. (b) Ground-truth map.

As mentioned before, the S^2RGF prototypes exhibit high computational complexity with respect to the kernel size, while the proposed DS^2RGF method could reduce it into a linear one. To verify this advantage, we measured their processing time and showed it as a function of the Gabor kernel size in Fig. 9. Clearly, DS^2RGF outperforms the corresponding prototype in terms of computational cost. The computational time of DS^2RGF increases slightly, whereas that of the prototype grows exponentially as the Gabor kernel size increases.

B. Data Set 2: Zaoyuan

The second hyperspectral image was collected by the Operational Modular Imaging Spectrometer (OMIS) over the Zaoyuan region, China, in 2001. The Zaoyuan scene contains 137 × 202 pixels and 80 spectral bands ranging from 0.4 to 1.7 μm, where the first 64 spectral bands are within the region of 0.4–1.1 μm and the last 16 within the region of 1.06–1.7 μm. The groundtruth used as a reference for classification contains eight land-cover classes. As shown in Fig. 10, the classes denoted by C1–C8 represent Vegetable, Grape, Dry vegetable, Pear, Corn, Terrace/Grass, Bush-Lespedeza, and Peach, respectively.

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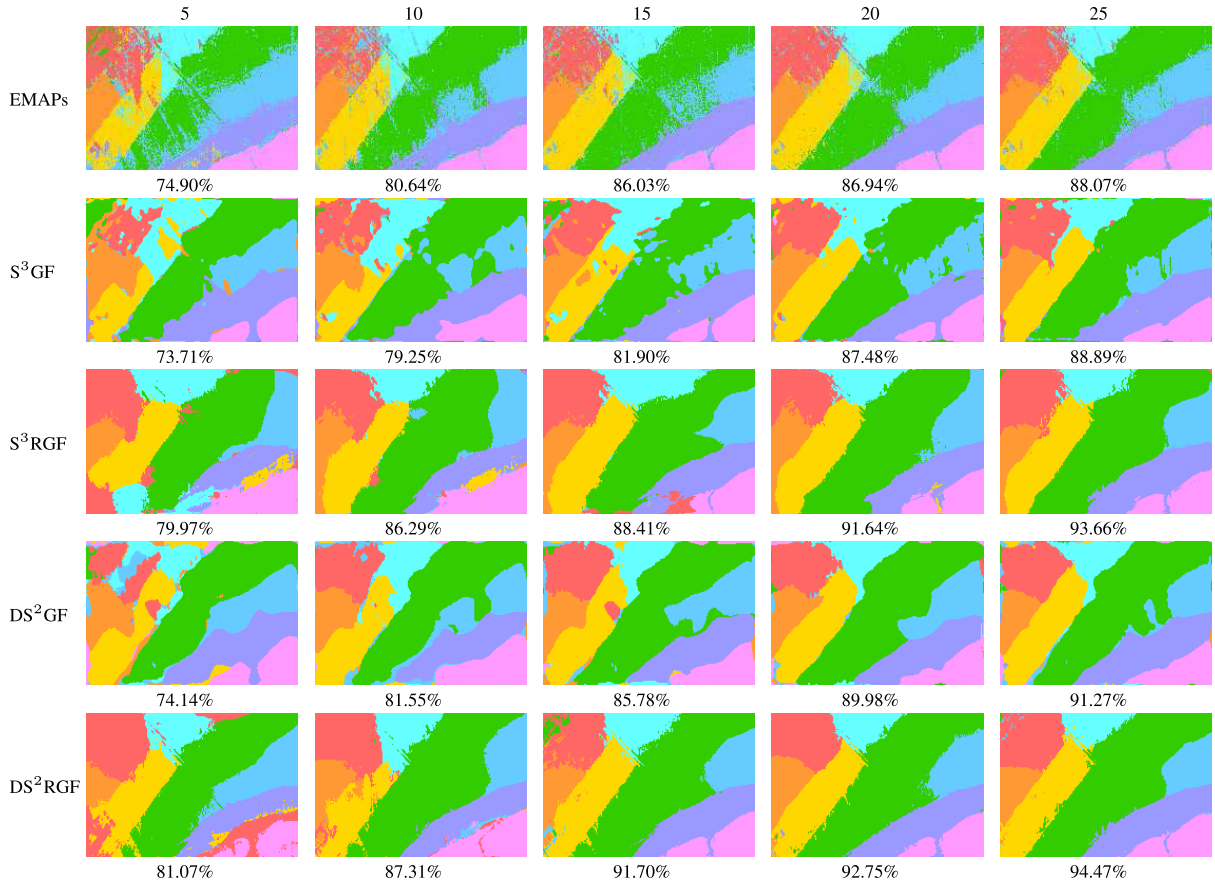


Fig. 11. Classification maps along with their corresponding OAs for the OMIS Zaoyuan scene, obtained from one of the 10 Monte Carlo runs by the MLR classifier with different numbers of training samples per class (corresponding to each column) on the considered features (corresponding to each row).

TABLE V

CAs (%), OAs (%), AAs (%) and κs (%) for the OMIS Zaoyuan Scene, Obtained From 10 Monte Carlo Runs by the MLR Classifier With 15 Training Samples Per Class

Class	♯Test	EMAPs	S^3GF	S^3RGF	DS^2GF	DS^2RGF
C1	2610	77.02	77.03	87.94	86.74	**89.38**
C2	1287	**95.59**	94.49	94.65	93.33	94.52
C3	3427	90.71	87.46	**94.05**	88.40	93.88
C4	10228	77.22	76.33	79.86	78.65	**86.16**
C5	1410	90.89	88.02	96.95	84.84	**97.77**
C6	1469	91.10	**94.61**	87.30	91.68	90.03
C7	1793	**98.90**	93.65	96.78	94.58	97.59
C8	1477	96.53	93.09	98.23	95.75	**98.39**
OA		84.66	82.79	87.51	86.88	**90.64**
AA		89.75	87.75	91.97	91.34	**93.47**
κ		80.74	78.36	84.21	83.47	**88.03**

First and foremost, we quantitatively evaluate the classification performance obtained for this scene in Table IV. Compared to the other feature extraction methods, the proposed DS^2RGF leads to a significant increase in terms of classification accuracies, especially with limited training samples. Take, for instance, the case of MLR and five training samples per class. DS^2RGF achieves an OA increase of around 3% in comparison to S^3RGF and, meanwhile, an increase of around 9% with regard to the other three. Therefore, it can be inferred that our proposed DS^2RGF method, with the inclusion of class boundary information and discriminative components, is able to greatly improve the quality of Gabor features. Furthermore, we list CAs, OAs, AAs, and κs by MLR with 15 training samples per class in Table V. In this scene, significant improvements can be observed in most of the classes.

We also illustrate the obtained classification maps in Fig. 11. It is remarkable that the classification maps obtained with the proposed DS^2RGF, i.e., those in the last row, look more unified and structured than the others. As expected, the regular shapes of fields fit well with the real scene, and less noises appear on the classification maps, which suggests very good performance of DS^2RGF in classification tasks.

Finally, we visualize their first principal components in Fig. 12. As observed, the features generated by DS^2RGF are more discriminative than the others. They are not only smooth from an intraclass level perspective but also contrasted and clear-cut from an interclass point of view. Those by S^3GF, DS^2GF, and even S^3RGF, are blurred to some degree, while the EMAP method is not that good when it comes to denoising as the Gabor filtering methods.

C. Data Set 3: Salinas

We conducted the third experiment on another AVIRIS data set over Salinas Valley, California, also available online. The Salinas scene is composed of 512 lines by 217 samples and characterized by a high spatial resolution of 3.7 m. Since our

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TABLE VI

OAs (%) ALONG WITH STANDARD DEVIATIONS FOR THE AVIRIS SALINAS SCENE, OBTAINED FROM 10 MONTE CARLO RUNS BY THE MLR AND SVM CLASSIFIERS WITH DIFFERENT NUMBERS OF TRAINING SAMPLES PER CLASS, RESPECTIVELY

Classifier	MLR					SVM				
♯Train/class	6	9	12	15	18	6	9	12	15	18
EMAPs	86.28±1.57	86.94±1.60	89.77±1.93	90.10±1.90	90.19±1.59	83.32±7.94	86.45±3.24	89.46±2.26	89.87±1.22	90.23±1.31
S^3GF	80.51±2.53	82.88±5.13	85.86±1.47	86.51±2.20	87.77±2.17	76.27±3.58	81.01±1.58	83.55±1.82	84.13±1.85	84.90±2.07
S^3RGF	88.61±3.47	94.78±1.57	95.12±1.85	96.69±2.01	97.10±1.14	83.42±2.91	91.12±2.13	93.87±2.18	95.28±2.42	95.87±0.99
DS^2GF	85.46±2.27	86.32±3.36	88.45±1.43	89.63±2.20	90.71±1.72	83.29±2.44	85.39±1.30	86.84±1.84	88.67±2.12	89.11±1.59
DS^2RGF	**93.28±1.84**	**96.13±1.46**	**96.37±1.78**	**96.75±2.31**	**98.15±1.46**	**90.21±2.78**	**94.54±2.05**	**95.86±2.23**	**96.74±2.36**	**97.02±1.74**

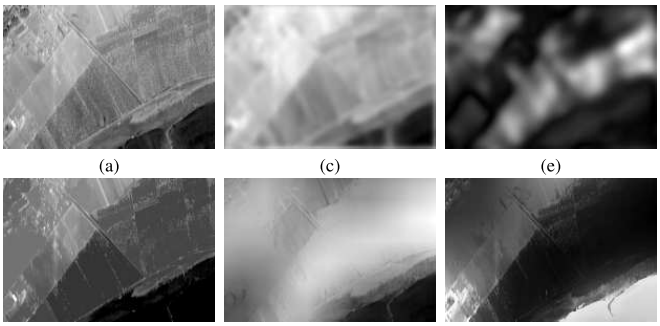


Fig. 12. First principal components obtained from different features for the OMIS Zaoyuan scene. (a) Original. (b) EMAPs. (c) S^3GF. (d) S^3RGF. (e) DS^2GF. (f) DS^2RGF.

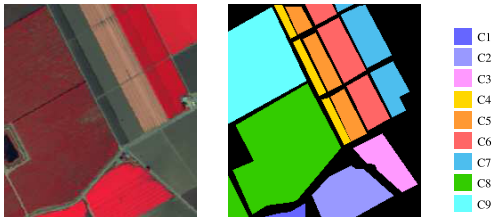


Fig. 13. AVIRIS Salinas scene. (a) False color map. (b) Ground-truth map.

focus is on the ability of classifiers to preserve strong edges to release the computational burden, we extracted the upper part containing abundant texture information for experimental evaluation. As a result, the used Salinas scene (see Fig. 13) comprises 240 × 217 samples and 204 spectral channels after moving 20 water absorption bands. Its groundtruth available contains nine classes, namely, Broccoli-green-weeds-1, Broccoli-green-weeds-2, Fallow, Fallow-rough-plow, Fallow-smooth, Stubble, Celery, Grapes-untrained, and Vineyard-untrained corresponding to the C1–C9, respectively, in Fig. 13.

Tables VI and VII list the OAs along with standard deviations versus the training size, and the CAs, OAs, AAs, and κs obtained by MLR with 12 training samples per class, respectively. The excellent performance of the proposed DS^2RGF can be observed from the two tables. That is, DS^2RGF outperforms the others with more accurate label assignments. Also, DS^2RGF achieved impressive results on the two biggest classes, i.e., C8 and C9, where the CAs are around 20% higher than EMAPs and S^2GF methods.

TABLE VII

CAs (%), OAs (%), AAs (%) AND κs (%) FOR THE AVIRIS SALINAS SCENE, OBTAINED FROM 10 MONTE CARLO RUNS BY THE MLR CLASSIFIER WITH 12 TRAINING SAMPLES PER CLASS

Class	♯Test	EMAPs	S^3GF	S^3RGF	DS^2GF	DS^2RGF
C1	272	**100.0**	97.06	99.93	98.64	99.17
C2	3388	**99.52**	97.13	97.40	98.99	97.90
C3	1964	**99.83**	93.33	97.80	99.49	98.17
C4	1382	99.35	96.11	98.74	94.76	**99.54**
C5	2666	**99.13**	95.33	96.73	97.27	98.30
C6	3947	**99.62**	99.09	97.04	98.08	98.55
C7	3567	**99.71**	95.66	95.88	98.77	98.00
C8	11129	76.53	72.19	92.03	74.42	**93.10**
C9	7256	86.91	81.71	95.20	86.93	**96.76**
OA		89.77	85.86	95.12	88.45	**96.37**
AA		95.62	91.95	96.75	94.15	**97.78**
κ		87.60	82.85	94.06	85.99	**95.59**

Fig. 14 shows the classification maps obtained with the considered methods. As can be seen, the classification maps obtained with DS^2RGF contain more smooth and detailed features than the others, followed by those with S^3RGF. On the one hand, the improvements for C8 and C9 can be again verified from these maps, where the leftmost part in the first, second, and fourth rows are affected by salt and pepper noises or deformed shapes, whereas those in the rest of the two rows are not. On the other hand, the paths between fields at the top-middle part of the scene shows up clearly only in the third and last rows. These observations indicate that the proposed S^2RGF methods exhibit a good potential to discriminate the classes while filtering HSIs.

Finally, we show the first principal components in Fig. 15. It can be observed that the features by DS^2RGF are of a higher quality than the others. The proposed DS^2RGF works well in terms of both denoising and deblurring. However, the EMAPs fail to remove noises as effectively as DS^2RGF. The features derived by S^3GF and DS^2GF are blurred. Besides, those by S^3RGF is also a little bit blurred for this scene, yet still look better than those by spectral–spatial Gabor filters.

D. Data Set 4: University of Houston

Urban environments usually include many natural and artificial ground fabrics, thus causing the appearance of complex spatial regions with high spectral variability [68]–[70]. Here, we extended our experiments to an urban scene given by the well-known and widely available University of Houston data set. This data set was acquired using the Compact

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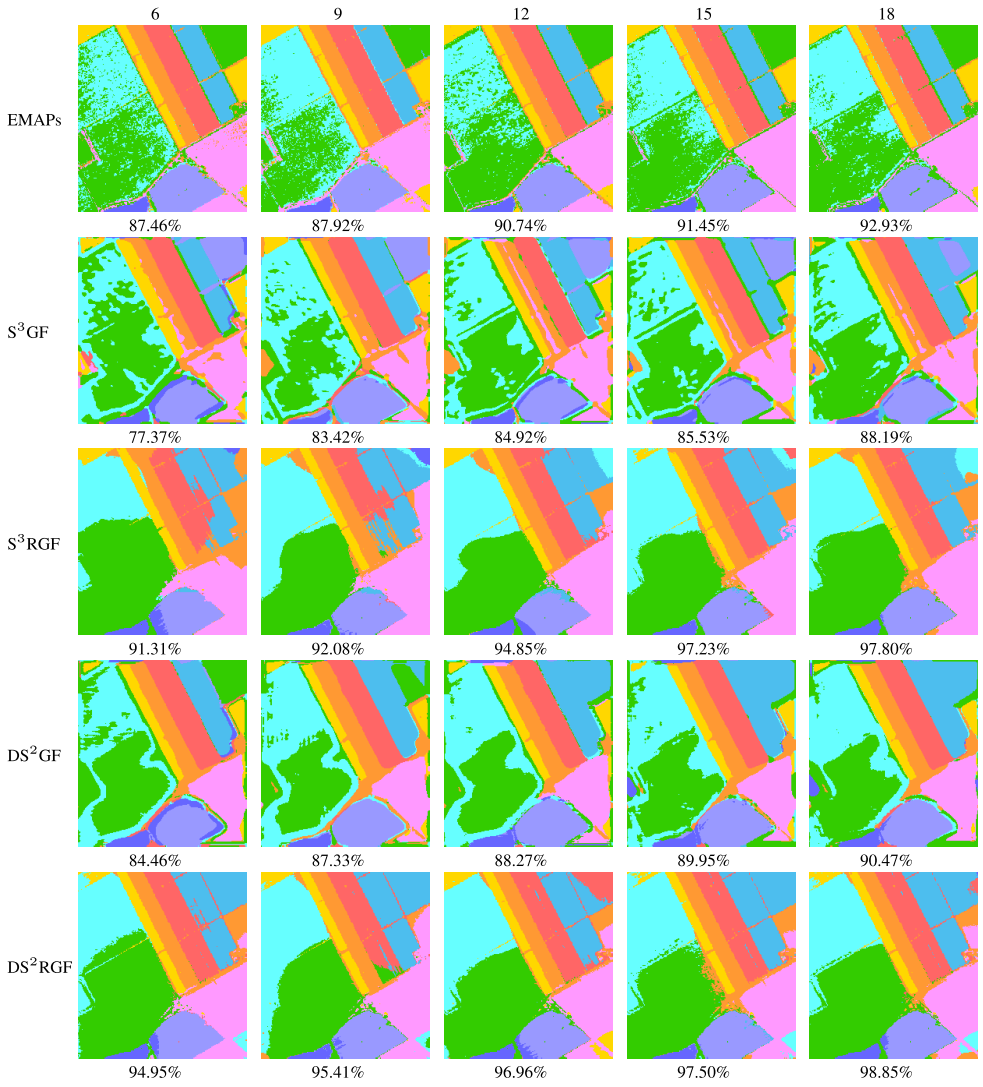


Fig. 14. Classification maps along with corresponding OAs for the AVIRIS Salinas scene, obtained from one of 10 Monte Carlo runs by the MLR classifier with different numbers of training samples per class (corresponding to each column) on the considered features (corresponding to each row).

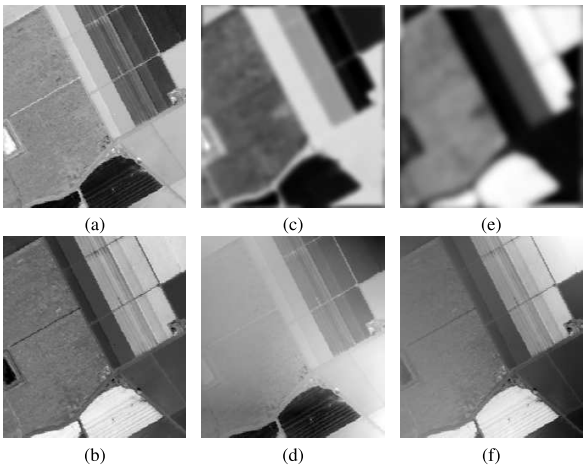


Fig. 15. First principal components obtained on different features for the AVIRIS Salinas scene. (a) Original. (b) EMAPs. (c) S^3GF. (d) S^3RGF. (e) DS^2GF. (f) DS^2RGF.

Airborne Spectrographic Imager from the ITRES company (ITRES-CASI 1500) over an urban area around the University of Houston campus, with 144 spectral bands ranging from 0.38 to 1.05 μm and a spatial resolution of 2.5 m. In our experiments, a subset covering 210 $\times$ 425 pixels was used, where eight classes are identified in the groundtruth with 612 training samples in total [see Fig. 16(a)–(c)].

Table VIII shows the obtained classification results, indicating the remarkable performance of our proposed S^2RGF methods. Furthermore, Fig. 16 gives a visual comparison of the considered feature extraction methods by using the MLR classifier. As shown, S^2RGF can generate better classification maps compared to the other methods, especially in the task of characterizing roads. As we can see, many roads are omitted by EMAPs and blurred by S^2GF methods, whereas DS^2RGF and S^3RGF preserve them better. These results verify the effectiveness of the proposed S^2RGF in urban remote sensing scenarios.

As shown in Tables I–VIII, the performance improvements of the proposed DS^2RGF with respect to S^3RGF are a little inconsistent on various data sets. The improvement on Indian Pines data set is not as much as those on the other three data sets. The reason for the phenomenon could be as follows. According to (14) and (18), the kernel of DS^2RGF is just

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TABLE VIII

CAs (%), OAs (%), AAs (%) AND κS (%) FOR THE UNIVERSITY OF HOUSTON SCENE, OBTAINED BY THE MLR AND SVM CLASSIFIERS WITH A FIXED TRAINING SET, RESPECTIVELY

Class	♯Train	♯Test	LORSAL					SVM				
			EMAPs	S^3GF	S^3RGF	DS^2GF	DS^2RGF	EMAP	S^3GF	S^3RGF	DS^2GF	DS^2RGF
Healthy Grass	45	72	**100.0**	88.89	83.33	80.56	76.39	**100.0**	65.28	70.83	94.44	**100.0**
Trees	59	422	99.29	99.53	**100.0**	96.68	**100.0**	98.10	93.36	98.34	**98.58**	96.45
Soil	25	71	100.0	91.55	**100.0**	98.59	**100.0**	**100.0**	**100.0**	**100.0**	95.77	**100.0**
Water	182	113	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	99.12	**100.0**	**100.0**	**100.0**	**100.0**
Commercial	76	151	96.69	96.69	**98.01**	95.36	**98.01**	92.05	96.03	96.69	**100.0**	98.01
Road	10	21	0.00	38.10	57.14	**100.0**	**100.0**	28.57	0.00	0.00	28.57	**95.24**
Parking Lot 1	57	213	**100.0**	86.38	84.04	**100.0**	**100.0**	95.77	**100.0**	98.59	99.06	**100.0**
Parking Lot 2	158	215	93.49	**98.60**	97.67	97.21	**98.60**	92.56	**100.0**	95.35	95.35	95.81
OA			96.64	94.84	95.07	96.71	**98.20**	95.23	93.74	94.76	96.87	**97.81**
AA			86.18	87.47	90.03	96.05	**96.63**	88.27	81.83	82.84	88.97	**98.19**
κ			95.83	93.59	93.88	95.92	**97.76**	94.09	92.20	93.46	96.11	**97.29**

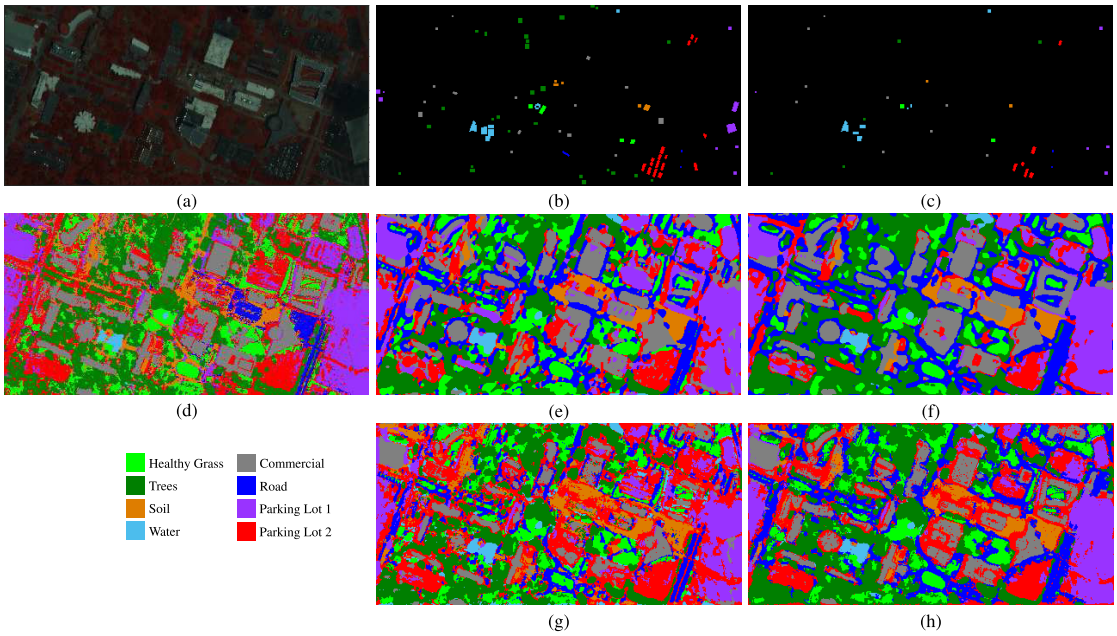


Fig. 16. Classification maps along with the corresponding OAs obtained by the MLR classifier for the University of Houston scene, where the false color map (with the selected RGB bands in brackets), the ground-truth, and the fixed training set are also displayed. (a) False color map [90,40,21]. (b) Ground truth. (c) Train. (d) EMAPs (96.64%). (e) S^3GF (94.84%). (f) DS^2GF (96.71%). (g) S^3RGF (95.07%). (h) DS^2RGF (98.20%).

one component of the kernel of S^3RGF, which is spatially low-pass. Thus, DS^2RGF tends to be more smooth an image than S^3RGF. On the other hand, Indian Pines is characteristic of higher intraclass smoothness than other data sets, especially Zaoyuan (can be observed in Figs. 3 and 10). In this situation, the potential of DS^2RGF to further smooth Indian Pines is limited, thus the performance improvement of DS^2RGF with respect to S^3RGF tends to be lower on this data set than on other data sets.

E. Parameter Settings

In practical applications, the optimal parameters vary on different hyperspectral data sets due to their different characteristics. There are three kinds of parameters that have to be set for spectral–spatial-range Gabor filtering methods, i.e., scale factors for the spatial domain, spectral domain, and range domain, respectively. Empirically, we found that the rural scenes could bear a larger scale in the spatial domain than the urban ones since the former ones are composed of the fields that are generally homogenous and of a large area. On the contrary, the urban scenes are featured with many complicated artifacts, where the delicate structure would be easily damaged by a large-scale filter. Also, the spatial scale of the bilateral Gabor filters can be set a bit larger than those of their traditional counterparts. This is expected since the bilateral versions take into consideration range domain variations, thus capable of resisting the oversmoothing phenomena around the spatial regional boundaries to some degree. As for the range scale, it is up to the intensity characteristics. On urban scenes, for example, most of the artifacts have very similar reflection properties. Therefore, a large range scale is usually not suitable for them.

In our experiments, we used cross-validation to determine the involved parameters, along with the aforementioned empirical searching considerations. In experiments, we found that the spatial scale factor is the most sensitive, and the range scale is less, and the spectral scale is the least. Accordingly, we employed an empirical searching. At first, the spectral scale

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was set to be 1. At first, the spectral scale is set to be 1. Then, the spatial scale and the range scale are jointly tuned with a grid search which includes two chief phases. First, we jointly search the range scale and the spatial scale in wide scope with large steps. Then, around the roughly determined pair of parameters, we find the optimal spatial scale and range scale with an alternative searching strategy in fine steps, i.e. search one given another and then search another with the first one fixed. Once both the range scale and the spatial scale are set, we in turn tune the spectral scale. The three kinds of scales can be alternatively tuned more than once for possibly better results. But such a fine-tuning comes at the expense of less time-consumption.

V. Conclusion

In this article, following a unified HSI filtering framework, we proposed a new concept of spectral–spatial-range Gabor filtering and then developed a discriminative spectral–spatial-range Gabor filtering method for HSI classification, which can not only extract discriminative spectral–spatial harmonic components but also relieve the feature deterioration caused by spatial regional boundaries. This novel Gabor filtering method is characterized by its ability to extract highly discriminative Gabor features and with a linear convolutional implementation that keeps its computational complexity linear. The proposed method was evaluated on four real HSI data sets, demonstrating excellent performance in all cases. Essentially, our spectral–spatial-range filtering method is based on a single-layer shallow learning, as opposed to deep learning methods involving multiple levels of representations and multiple layers of processing units [71]–[73]. Although the work in this article is limited to the single-layer learning case, we plan to design model prior-based deep neural networks that integrate our new Gabor filtering scheme in future developments.

References

[1] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

[2] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 45–54, Jan. 2014.

[3] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[4] A. Plaza *et al.*, "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, no. 1, pp. S110–S122, 2009.

[5] A. Zare and K. Ho, "Endmember variability in hyperspectral analysis: Addressing spectral variability during spectral unmixing," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 95–104, Jan. 2014.

[6] G. Hughes, "On the mean accuracy of statistical pattern recognizers," *IEEE Trans. Inf. Theory*, vol. IT-14, no. 1, pp. 55–63, Jan. 1968.

[7] K. Fukunaga, *Introduction to Statistical Pattern Recognition*. San Diego, CA, USA: Academic, 1990, ch. 3.

[8] M. J. Khan, H. S. Khan, A. Yousaf, K. Khurshid, and A. Abbas, "Modern trends in hyperspectral image analysis: A review," *IEEE Access*, vol. 6, pp. 14118–14129, 2018.

[9] R. Kawakami, Y. Matsushita, J. Wright, M. Ben-Ezra, Y.-W. Tai, and K. Ikeuchi, "High-resolution hyperspectral imaging via matrix factorization," in *Proc. CVPR*, Jun. 2011, pp. 2329–2336.

[10] Z. Du, M. K. Jeong, and S. G. Kong, "Band selection of hyperspectral images for automatic detection of poultry skin tumors," *IEEE Trans. Autom. Sci. Eng.*, vol. 4, no. 3, pp. 332–339, Jul. 2007.

[11] C.-H. Li, B.-C. Kuo, C.-T. Lin, and C.-S. Huang, "A spatial–contextual support vector machine for remotely sensed image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 784–799, Mar. 2012.

[12] X. Guo, X. Huang, L. Zhang, L. Zhang, A. Plaza, and J. A. Benediktsson, "Support tensor machines for classification of hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3248–3264, Jun. 2016.

[13] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.

[14] Z. Alasvand, M. Naderan, and G. Akbarizadeh, "Superpixel-based feature learning for joint sparse representation of hyperspectral images," in *Proc. 3rd Int. Conf. Pattern Recognit. Image Anal. (IPRIA)*, Apr. 2017, pp. 156–159.

[15] Y. Tarabalka, J. Chanussot, and J. Benediktsson, "Segmentation and classification of hyperspectral images using watershed transformation," *Pattern Recognit.*, vol. 43, no. 7, pp. 2367–2379, Jul. 2010.

[16] Y. Tarabalka, J. Chanussot, and J. A. Benediktsson, "Segmentation and classification of hyperspectral images using minimum spanning forest grown from automatically selected markers," *IEEE Trans. Syst. Man, Cybern. B, Cybern.*, vol. 40, no. 5, pp. 1267–1279, Oct. 2010.

[17] X. Jia, B.-C. Kuo, and M. M. Crawford, "Feature mining for hyperspectral image classification," *Proc. IEEE*, vol. 101, no. 3, pp. 676–697, Mar. 2013.

[18] T. M. Cover and J. A. Thomas, *Elements of Information Theory*. New York, NY, USA: Wiley, 1991, ch. 11.

[19] L. He, Q. Pan, W. Di, and Y. Li, "Anomaly detection in hyperspectral imagery based on maximum entropy and nonparametric estimation," *Pattern Recognit. Lett.*, vol. 29, no. 9, pp. 1392–1403, Jul. 2008.

[20] S. M. Kay, *Fundamentals of Statistical Signal Processing: Estimation Theory*, vol. 1. Upper Saddle River, NJ, USA: Prentice-Hall, 1993, ch. 3.

[21] L. L. Scharf, *Statistical Signal Processing: Detection, Estimation, and Time Series Analysis*. Reading, MA, USA: Addison-Wesley, 2006, ch. 6.

[22] S. M. Kay, *Fundamentals of Statistical Signal Processing: Detection Theory*, vol. 2. Upper Saddle River, NJ, USA: Prentice-Hall, 1993, ch. 5.

[23] R. O. Duda, P. E. Hart, and D. G. Stork, *Pattern Classification*. 2nd ed. New York, NY, USA: Wiley, 2001, ch. 2.

[24] C. Liu, L. He, Z. Li, and J. Li, "Feature-driven active learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 341–354, Jan. 2018.

[25] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[26] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[27] N. S. Altman, "An introduction to kernel and nearest-neighbor nonparametric regression," *Amer. Statist.*, vol. 46, no. 3, pp. 175–185, Aug. 1992.

[28] L. He, Y. Li, X. Li, and W. Wu, "Spectral–spatial classification of hyperspectral images via spatial translation-invariant wavelet-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2696–2712, Nov. 2015.

[29] J. Benediktsson, J. Palmason, and J. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[30] M. Dalla Mura, J. Atli Benediktsson, B. Waske, and L. Bruzzone, "Extended profiles with morphological attribute filters for the analysis of hyperspectral data," *Int. J. Remote Sens.*, vol. 31, no. 22, pp. 5975–5991, Dec. 2010.

[31] A. Plaza, P. Martinez, R. Perez, and J. Plaza, "A new approach to mixed pixel classification of hyperspectral imagery based on extended morphological profiles," *Pattern Recognit.*, vol. 37, no. 6, pp. 1097–1116, Jun. 2004.

[32] A. Plaza, P. Martinez, J. Plaza, and R. Perez, "Dimensionality reduction and classification of hyperspectral image data using sequences of extended morphological transformations," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 466–479, Mar. 2005.

[33] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2008.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 14:22:44 UTC from IEEE Xplore. Restrictions apply.

[34] X. Xu, J. Li, X. Huang, M. D. Mura, and A. Plaza, "Multiple morphological component analysis based decomposition for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 3083–3102, May 2016.
[35] D. Gabor, "Theory of communications," *J. Inst. Electr. Eng.*, vol. 93, pp. 429–457, 1946.
[36] B. Kisacanin, *Embedded Computer Vision*. London, U.K.: Springer-Verlag, 2009, ch. 13.
[37] L. He, J. Li, A. Plaza, and Y. Li, "Discriminative low-rank Gabor filtering for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1381–1395, Mar. 2017.
[38] T. C. Bau, S. Sarkar, and G. Healey, "Hyperspectral region classification using a three-dimensional Gabor filterbank," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 9, pp. 3457–3464, Sep. 2010.
[39] L. Shen and S. Jia, "Three-dimensional Gabor wavelets for pixel-based hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 5039–5046, Dec. 2011.
[40] S. Jia, L. Shen, and Q. Li, "Gabor feature-based collaborative representation for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 1118–1129, Feb. 2015.
[41] G. Akbarizadeh, "A new statistical-based kurtosis wavelet energy feature for texture recognition of SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4358–4368, Nov. 2012.
[42] Z. Tirandaz and G. Akbarizadeh, "A two-phase algorithm based on kurtosis curvelet energy and unsupervised spectral regression for segmentation of SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 3, pp. 1244–1264, Mar. 2016.
[43] J.-K. Kamarainen, V. Kyrki, and H. Kalviainen, "Invariance properties of Gabor filter-based features-overview and applications," *IEEE Trans. Image Process.*, vol. 15, no. 5, pp. 1088–1099, May 2006.
[44] T. Yun and L. Guan, "Human emotional state recognition using real 3D visual features from Gabor library," *Pattern Recognit.*, vol. 46, no. 2, pp. 529–538, Feb. 2013.
[45] W.-S. Chen, W. Wang, J.-W. Yang, and Y. Y. Tang, "supervised regularization locality-preserving projection method for face recognition," *Int. J. Wavelets, Multiresolution Inf. Process.*, vol. 10, no. 6, Nov. 2012, Art. no. 1250053.
[46] A. Bovik, M. Clark, and W. Geisler, "Multichannel texture analysis using localized spatial filters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 12, no. 1, pp. 55–73, Jan. 1990.
[47] F. Bianconi and A. Fernández, "Evaluation of the effects of Gabor filter parameters on texture classification," *Pattern Recognit.*, vol. 40, no. 12, pp. 3325–3335, Dec. 2007.
[48] J. F. Khan, R. R. Adhami, and S. M. Bhuiyan, "A customized Gabor filter for unsupervised color image segmentation," *Image Vis. Comput.*, vol. 27, no. 4, pp. 489–501, Mar. 2009.
[49] J. Yi, C. Lee, and J. Kim, "Online signature verification using temporal shift estimated by the phase of Gabor filter," *IEEE Trans. Signal Process.*, vol. 53, no. 2, pp. 776–783, Feb. 2005.
[50] T. Chen, X. Wang, S. Chung, D. Metaxas, and L. Axel, "Automated 3D motion tracking using Gabor filter bank, robust point matching, and deformable models," *IEEE Trans. Med. Imag.*, vol. 29, no. 1, pp. 1–11, Jan. 2010.
[51] J. G. Daugman, "Uncertainty relation for resolution in space, spatial frequency, and orientation optimized by two-dimensional visual cortical filters," *J. Opt. Soc. Amer. A, Opt. Image Sci.*, vol. 2, no. 7, p. 1160, Jul. 1985.
[52] T. Sing Lee, "Image representation using 2D Gabor wavelets," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 18, no. 10, pp. 959–971, Oct. 1996.
[53] S. Quian and D. Chen, *Joint Time-Frequency Analysis: Methods and Applications*. New York, NY, USA: Prentince–Hall, 1996.
[54] P. Milanfar, "A tour of modern image filtering: New insights and methods, both practical and theoretical," *IEEE Signal Process. Mag.*, vol. 30, no. 1, pp. 106–128, Jan. 2013.
[55] Z. Wang and A. Bovik, "Mean squared error: Love it or leave it? A new look at signal fidelity measures," *IEEE Signal Process. Mag.*, vol. 26, no. 1, pp. 98–117, Jan. 2009.
[56] L. Pizarro, P. Mrázek, S. Didas, S. Grewenig, and J. Weickert, "Generalised nonlocal image smoothing," *Int. J. Comput. Vis.*, vol. 90, no. 1, pp. 62–87, Oct. 2010.
[57] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev.*, vol. 51, no. 3, pp. 455–500, Aug. 2009.
[58] S. Paris and F. Durand, "A fast approximation of the bilateral filter using a signal processing approach," *Int. J. Comput. Vis.*, vol. 81, no. 1, pp. 24–52, Jan. 2009.
[59] L. Xiao and L. Deng, "A geometric perspective of large-margin training of Gaussian models," *IEEE Signal Process. Mag.*, vol. 27, no. 6, pp. 118–123, Nov. 2010.
[60] A. V. Oppenheim, A. S. Willsky, and S. H. Nawab, *Signals and Systems*, 2nd ed. Englewood Cliffs, NJ, USA: Prentice-Hall, 1996, ch. 7.
[61] J. Li, P. R. Marpu, A. Plaza, J. M. Bioucas-Dias, and J. A. Benediktsson, "Generalized composite kernel framework for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 9, pp. 4816–4829, Sep. 2013.
[62] M. Khodadadzadeh, J. Li, A. Plaza, H. Ghassemian, J. M. Bioucas-Dias, and X. Li, "Spectral–spatial classification of hyperspectral data using local and global probabilities for mixed pixel characterization," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 10, pp. 6298–6314, Oct. 2014.
[63] J. Li, H. Zhang, and L. Zhang, "Efficient superpixel-level multitask joint sparse representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5338–5351, Oct. 2015.
[64] C. Pan, X. Gao, Y. Wang, and J. Li, "Markov random fields integrating adaptive interclass-pair penalty and spectral similarity for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2520–2534, May 2019.
[65] S. Li, T. Lu, L. Fang, X. Jia, and J. A. Benediktsson, "Probabilistic fusion of pixel-level and superpixel-level hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7416–7430, Dec. 2016.
[66] J. Hu, Z. He, J. Li, L. He, and Y. Wang, "3D-Gabor inspired multiview active learning for spectral–spatial hyperspectral image classification," *Remote Sens.*, vol. 10, no. 7, p. 1070, Jul. 2018.
[67] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Hyperspectral image segmentation using a new Bayesian approach with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3947–3960, Oct. 2011.
[68] P. Hardin and A. Hardin, "Hyperspectral remote sensing of urban areas," *Geogr. Compass*, vol. 7, no. 1, pp. 7–21, Jan. 2013.
[69] J. Franke, D. A. Roberts, K. Halligan, and G. Menz, "Hierarchical multiple endmember spectral mixture analysis (MESMA) of hyperspectral imagery for urban environments," *Remote Sens. Environ.*, vol. 113, no. 8, pp. 1712–1723, Aug. 2009.
[70] I. Meganem, P. Deliot, X. Briottet, Y. Deville, and S. Hosseini, "Linear–quadratic mixing model for reflectances in urban environments," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 544–558, Jan. 2014.
[71] Q. Gao, S. Lim, and X. Jia, "Hyperspectral image classification using convolutional neural networks and multiple feature learning," *Remote Sens.*, vol. 10, no. 2, p. 299, Feb. 2018.
[72] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.
[73] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.



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